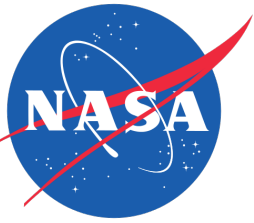


CLDC Industry Update May 2025 Charts



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CLDS Requirements and Standards Updates



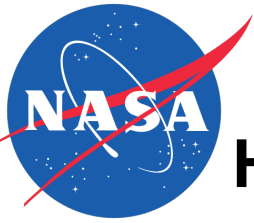
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CLDP-REQ-1130 Rev. A is being developed for Draft RFP



- **Baseline of CLDP-REQ-1130, Requirements and Standards was provided to industry on January 23, 2025**
- **Rev. A is being developed to incorporate feedback on the initial baseline and other open items**
 - Adjustments as a result of ASM-2 in December, 2024
 - Internal Independent Review (177 comments)
 - Feedback from Industry (1,111 comments) on CLDP-REQ-1130 Baseline from the Requirements and Safety TIM part 2 held 1/28/25
 - Forward Work items identified in the Baseline CR plus additional items assigned later (85 items)
- **Most significant changes in work:**
 - Utilization requirements are being reworked based on Agency and Industry feedback, including development of a new Utilization Standard
 - Alignment of requirements with Draft RFP drove removal of some requirements from 1130
 - Descoping of fixed fire suppression requirements from the Destination
 - New consolidated standards for electromagnetic effects
 - Communication requirements clarifications and refinement



Heritage Visiting Vehicle Transportation Interface Requirements



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Heritage Visiting Vehicle (VV) Transportation Background



- **NASA and the commercial transportation sector have been working together since 2006 for crew and cargo transportation services to and from the International Space Station (ISS). These existing vehicles are what is referred to as the heritage transportation that could be offered as Government Furnished Flights (GFF)**
 - SpaceX's Dragon, docking vehicles with return to earth capability and re-use
 - Boeing's Starliner, docking vehicle with return to earth capability and re-use
 - Sierra Space's Dreamchaser, berthing vehicle with return to earth capability and re-use
 - Northrup Grumman's Cygnus, berthing vehicle with no return to capability and disposed of during re-entry
- **The heritage vehicles are currently designed, built and operated based on the Space Station Program (SSP) 50808 Interface Requirement Document (IRD) which provides the requirements for the transportation vehicles to safely and successfully operate within the ISS's Approach Ellipsoid (AE) during Rendezvous Proximity Operations and Docking (RPOD) phases of the missions and for attached phases.**
- **ISS specifications required some unique system interfaces be built into the heritage vehicles that may not be compatible with CLDs. Additional discussion with industry is needed on the following items:**
 - Altitude and inclination
 - C2V2 for ship to ship communications
 - TDRS for space to ground communications
 - MIL-STD-1553 for docked communications and commanding
 - 10/100 Ethernet for docked network communications
 - Power needs
 - Jet Firing/Pulse Train characteristics



Heritage VV Transportation Requirements Overview



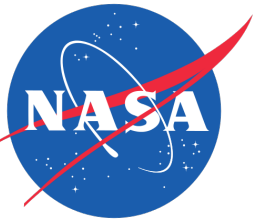
- **In order to ensure compatibility between CLDs and Heritage Visiting Vehicles provided as part of NASA government provided transportation, NASA is developing a heritage visiting vehicle requirements document planned to be released with the Phase 2 Draft RFP**
- **This will cover interface functions between the Destination and Transportation vehicle, such as:**
 - Structural/Mechanical Attachments
 - Crew/Cargo Passageway access
 - Guidance, Navigation and Control (GN&C) aids for rendezvous and docking
 - Visiting Vehicle Power needs
 - Data and Commanding
 - Intermodule air exchange
 - Vestibule Pressurization/Depressurization



- **Heritage Visiting Vehicle requirements for GFF docking transportation vehicles are based on CLDP-STD-1101, CLDP Common Docking System Standard**
 - Berthing and robotics requirements will be in an appendix as an optional capability
 - List of current heritage transportation waivers against SSP 50808 are in an appendix of the heritage Visiting Vehicle requirements document
- **The heritage visiting vehicle requirements document will be invoked as applicable in CLDP-REQ-1130, Requirements and Standards for the CLDP**
 - Some requirements from SSP 50808 were written into CLDP-REQ-1130 so there is currently some overlap of requirements in both documents which will be resolved prior to Final RFP
- **NASA envisions CLD specific IRDs will be required between CLD/VV to document CLD specific requirements, for end to end transportation and may also be used for compliance to these requirements.**



- **Release Heritage Visiting Vehicle requirements document NLT Draft RFP for industry review**
- **A number of TBDs and TBRs will remain in the initial version with goal to resolve by Final RFP**
 - TBDs against SSP applicable documents
 - TBRs against requirements that are ISS specific and will need CLD specifics or NASA standards
 - TBDs against ISS legacy systems C2V2, TDRS, 10/100 Ethernet, MIL-STD-1553, etc
- **Industry feedback and comments are needed to aid NASA in finding a balance between maintaining flexibility in CLD Design trades and minimizing changes driving heritage visiting vehicle redesign and recertification**



Safety & Mission Assurance Updates



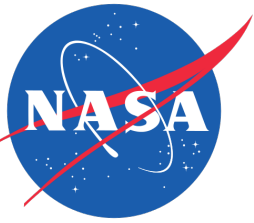
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Safety & Mission Assurance (S&MA) Updates



- **Based on industry feedback, NASA is updating CLDP-REQ-3102, Hazard Analysis and Safety Requirements Process, as CLDP-STD-3102, Hazard Analysis and Safety Process Standard, as well as to clarify some areas, provide additional content, and allow for a Provider led, innovative approach**
 - New sections on ground safety process (example from ISS) and hazardous material control strategy
 - Design for Minimum Risk (DFMR) guidance list added (example from ISS) and list of technical data required per phase to support hazard analysis
 - Expanded computer safety and emergency response sections
 - Clarification and additional details surrounding Joint Safety Reviews
 - Additional requirement in CLDP-REQ-1130 stating the need to meet the intent of CLDP-STD-3102
- **Although current NASA safety process has been provided as an example, the CLD Provider may propose how to meet the safety process requirements as part of their proposal and DRD submissions.**
 - Final agreements will be worked post contract award with details prior to first phased safety review
 - As part of this process, NASA/CLD Provider will establish clear roles, responsibilities and framework among all stakeholders
- **Forward Work: NASA is still working updates associated with industry comment and questions related to roles and responsibilities and updates associated with government provided transportation**



Proposed Medical Requirements



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- **The CLD Provider is responsible for developing and establishing medical standards and evaluation requirements for medical & behavioral qualification of CLD crew.**
- **Based on industry feedback and NASA's recent experience with Private Astronaut Missions (PAMs), NASA will levy a targeted set of minimum medical & behavioral standards and medical evaluation requirements that the Provider must include in their medical certification standards.**
 - NASA requirements for medical and behavioral qualification of CLD crew are intended to ensure crew safety, enable mission success, and support broader access to space.
 - The intent is to establish NASA's expectations for acceptable medical standards and to convey NASA's experience in spaceflight medical qualification.
 - Medical standards are intended to establish acceptable risk thresholds and identify conditions or findings posing elevated risk.
 - The goal is to adequately investigate, disposition, and mitigate medical and behavioral risks to ensure crew safety and mission success rather than to disqualify individuals.



Medical & Behavioral Qualifications



- **Medical and behavioral qualifications ensure that crews' health status poses an acceptable medical and psychiatric risk to crew safety and NASA mission success**
 - Medical and behavioral standards identify those conditions or findings that may pose an elevated risk to crew safety or negatively impact mission success and may require further investigation and risk mitigation.
 - NASA standards are NOT disqualifying; instead, they establish acceptable risk thresholds the exceeding of which requires further assessment and risk mitigation.
 - This approach enables insight into medical risk posed by existing or yet unidentified medical conditions and identification of appropriate monitoring, interventions, or countermeasures to mitigate risk prior to flight.
 - Thresholds for acceptable risk will be tailored based on mission duration and crew roles and responsibilities
- **The Commercial Provider will be responsible for establishing medical standards (conditions posing elevated risk) and medical evaluation requirements to assess acceptable medical & behavioral risk.**
 - The Commercial Provider's medical standards and evaluation requirements must include NASA-specified minimums
 - The CLD Provider may establish additional standards or evaluation requirements for mission certification beyond those specified by NASA.
 - Any standards or medical evaluation requirements exceeding NASA minimums levied on USG crew will be reviewed and approved by NASA.
 - NASA internal crew selection standards exceed the CLD minimum standards noted above.



Psychological Suitability



- **Psychological suitability encompasses critical behavioral competencies (or attributes) such as problem-solving, teamwork, leadership/followership, self-regulation, resilience, and adaptability that enable safe and successful execution of long duration missions.**
- **The Commercial Provider is responsible for ensuring crews are assessed and deemed psychologically suitable for the mission.**
 - Psychological suitability assessment is intended to enable identification of risk posed by lack of critical behavioral competencies, and to implement appropriate mitigations prior to flight.
 - The Provider is responsible for conducting a multi-modal assessment consisting of psychodiagnostic testing, psychiatric/psychological interview by a specialist, and observation of operational simulations to assess behavioral competencies.
 - At a minimum, results of the assessment should indicate that the individual possesses sufficient levels of critical attributes (competencies) to:
 - Function as a team member or crewmate in an operational setting
 - Regulate and manage emotions, behaviors, and impulses
 - Tolerate high levels of stress and effectively cope with and adapt to stressful conditions
 - Apprehend the impact of one's emotions or behaviors on others
 - Live cooperatively with others in austere and confined conditions and is tolerant of others' differences, and aware and respectful of others' personal needs and boundaries in those living conditions
 - Any deficiencies identified during psychological suitability assessment are to be addressed via mission selection or training.



Medical & Behavioral Qualification Process



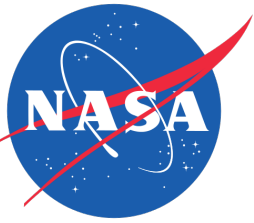
- **The CLD Provider will establish a process for medical and behavioral qualification of USG and non-USG crew**
 - Process will include medical history, physical exam, laboratory testing, imaging and procedures as specified in Provider mission medical and behavioral selection standards.
 - NASA will perform mission medical & behavioral qualification for USG crews based on NASA and CLD specific requirements.
 - Timeline for execution of mission medical selection testing and certification needs to ensure crew are medically & behaviorally certified prior to start of hazardous training and that crew medical certification remains valid through return to Earth.
 - Re-certification maybe be required to ensure that medical and behavioral certification is valid through landing



- **The CLD Provider will establish a review board to evaluate medical and behavioral testing, disposition risk & determine mitigations that meet the following guidelines:**
 - Ensure medical and behavioral certification is performed and approved by aerospace medicine board certified or eligible physicians and with training in assessment of medical and behavioral fitness for duty for hazardous ground training and spaceflight
 - Identify findings from crew medical and behavioral assessments that do not meet mission medical selection standards and/or pose an elevated risk to the USG crew or to the NASA mission.
 - Ensure conditions that do not meet mission medical selection standards and/or pose an elevated risk to USG crew or to the NASA mission are mitigated and that residual risk is deemed acceptable to qualify for spaceflight duties
 - Assess impacts to medical certification resulting from changes in crew health status that arise after crew medical certification and that are in violation of mission medical selection standards and/or pose an elevated risk to USG crew or to the NASA mission.
 - Include NASA as a participant in the assessment and disposition of medical, behavioral, and psychological suitability risk of non-USG crew to assess propagated risk to USG crew or to the NASA mission



- **NASA's requirements for medical and behavioral qualifications, including psychological suitability and a minimum set of medical standards and evaluation requirements will be released with the draft RFP for industry feedback.**
- **NASA is assessing whether it can offer the expertise of its Aerospace Medicine Board as a service to Providers to augment Provider aerospace medicine expertise in conducting medical certification for training and spaceflight.**
 - If available, this information will be released as part of Draft RFP and offered as Government Services to be negotiated as part of potential future Government Task Agreements (GTAs) or Reimbursable Space Act Agreements (RSAA).



CLDS Utilization Requirement Updates



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Utilization Requirement Updates



- **Based on internal and industry feedback, NASA is moving most Utilization requirements out of CLDP-REQ-1130 into a new Utilization Standard CLDP-STD-1106**
 - Clarification provided April 2, 2025, through SAM.gov regarding CLDS Requirements / Assumptions to give Providers flexibility in range of Baseline Lab Capabilities (BLC) and sequencing of implementation
 - Utilization capabilities are not all required at Initial Operating Capability (IOC) but can be implemented in phases (Min IOC, Continuous Crew (CC), Full Operating Capability (FOC))
 - BLC capabilities organized into Tiers to align with expected science needs and reduced footprint to allow Providers to plan maturation/evolution path for phased groupings of BLC capabilities
- **CLDP-REQ-1130 Section 3.6/4.6 are being updated to:**
 - Limit scope to define resources needed at utilization interfaces
 - Move specific BLC requirements from CLDP-REQ-1130 into a new standard CLDP-STD-1106 CLDP BLC Standard
 - Incorporate 1 new ‘shall’ statement to levy CLDP-STD-1106 as a “meet the intent” standard
- **CLDP-STD-1106 allows Providers to meet the intent of the requirements groupings**
 - Clarify standards could be met by a combination of hardware (ex: Microscope related BLC requirements)
 - Provide information related to BLC items or variations that are not currently in CLDP-REQ-1130 that were targeted for Rev A (ex: Edge Computing Device)
 - Include an Appendix in CLDP-STD-1106 for reference with the power, thermal, and data usage rates for the current BLC facilities on ISS



Restructuring of Utilization Requirements

- CLDP-REQ-1130 scope will be narrowed to provide general payload requirements that may impact the design and development of Destination or Transportation systems for broad use-cases and scenarios

CLDP-REQ-1130

- Pressurized Payload Resources
- Unpressurized Payload Resources
- Data Management and Communication
- Microgravity Environment
- Environmental Monitoring
- “Meets the intent of CLDP-STD-1106” requirement

- CLDP-STD-1106 will define specific payload requirements that may have many potential implementation solutions for limited use-cases and scenarios

CLDP-STD-1106

- | | |
|------------------------|---------------------------------------|
| • Conditioned Stowage | • Analytical Tools |
| • Small-Arm Centrifuge | • Lab Equipment |
| • Ultrasound | • Edge Computing Facility |
| • Glovebox | • Associated Supplies and Consumables |
| • Workbench | • Mass Measurement Device |
| • Microscope | • Pharmacy Locker |
| • Wetlab | • Medium-Arm Centrifuge |
| • Bio Sequencer | • Plant Research Facility |
| • Cell/Tissue Facility | • Small area/Organism area |

Appendix will include power, thermal, and data usage rates for current NASA designed BLC systems for reference



Requirements expected to remain in CLDP-REQ-1130



Current Section 3.6.5 Pressurized Payload Resources

R.CLDS.612	PRESSURIZED PAYLOAD MECHANICAL ATTACHMENT
R.CLDS.614	PRESSURIZED PAYLOAD POWER
R.CLDS.615	PRESSURIZED PAYLOAD VOLTAGE
R.CLDS.616	PRESSURIZED PAYLOAD THERMAL COOLING
R.CLDS.617	PRESSURIZED PAYLOAD VACUUM EXHAUST
R.CLDS.618	PRESSURIZED PAYLOAD VACUUM RESOURCE
R.CLDS.619	PRESSURIZED PAYLOAD GAS RESOURCE
R.CLDS.693	POWER INVERTER AND CONVERTER

Current Section 3.6.6 Unpressurized Payload Resources

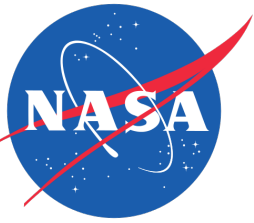
R.CLDS.621	UNPRESSURIZED PAYLOAD MECHANICAL ATTACHMENT
R.CLDS.622	UNPRESSURIZED PAYLOAD MASS AND VOLUME
R.CLDS.623	UNPRESSURIZED PAYLOAD POWER
R.CLDS.624	UNPRESSURIZED PAYLOAD VOLTAGE
R.CLDS.625	UNPRESSURIZED PAYLOAD THERMAL COOLING
R.CLDS.626	UNPRESSURIZED PAYLOAD SURVIVAL POWER DURING TRANSLATION
R.CLDS.628	UNPRESSURIZED PAYLOAD INSTALLATION AND REMOVAL CAPABILITY
R.CLDS.679	UNPRESSURIZED PAYLOAD DUAL-POWER CHANNEL

Current Section 3.6.7 Data Management and Communication

R.CLDS.474	PAYLOAD C&DH
R.CLDS.476	PAYLOAD TELEMETRY MONITORING
R.CLDS.477	PAYLOAD DATA RATES
R.CLDS.478	PAYLOAD DATA LATENCY
R.CLDS.479	PAYLOAD DATA STORAGE
R.CLDS.480	PAYLOAD DATA TRANSFER
R.CLDS.481	PAYLOAD DATA RECORDING
R.CLDS.620	PAYLOAD SOFTWARE INTERFACE

Current Section 3.6.9 Environmental Monitoring

R.CLDS.532	ATMOSPHERIC MONITORING
R.CLDS.533	RADIATION MONITORING
R.CLDS.534	MICROGRAVITY MONITORING
R.CLDS.675	ELECTROMAGNETIC INTERFERENCE MONITORING
R.CLDS.676	SPACECRAFT POINTING AND ATTITUDE MONITORING
R.CLDS.677	CONTAMINATION MONITORING
R.CLDS.678	GLOBAL POSITIONING MONITORING
R.CLDS.727	DISTURBANCE MONITORING
R.CLDS.728	MICROGRAVITY ENVIRONMENT



Utilization Approach Status



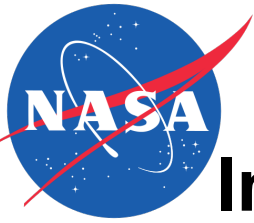
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Utilization Approach Status



- **CLDP has held multiple TIMs which discussed the Utilization Approach**
 - Requirements and Utilization Operations TIM - June 2024
 - Requirements and Safety TIM – January 2025
- **The Overall Utilization Approach provided in past RFIs/TIMs has not changed.**
- **NASA will provide as part of the Draft RFP, CLDP-PLN-3017, ‘Commercial Low-Earth Orbit Development Program Utilization Integration Plan’ (aka, UIP)**
 - This document details expected roles and responsibilities, tasks, and milestones between NASA and industry
- **A Commercial Provider Utilization Integration Plan will be required to outline the Commercial Provider approach for completing utilization integration activities as part of contract proposal.**
 - Final integration roles and responsibilities, interfaces and details will be jointly agreed to post contract award.



Initial Operating Capability (IOC) / Continuous Crew (CC) / Full Operating Capability (FOC) Definitions



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Clarifications on CLDS Requirements / Assumptions



- **CLDP-REQ-1130 Requirements were intended to be applicable at Full Operating Capability (FOC), not Initial Operating Capability (IOC)**
 - IOC by Dec 2029, Continuous Crew (CC) by Dec 2030, FOC by Dec 2031
 - A significant majority of Baseline Laboratory Capabilities (BLC) are commercially available now (ex: COTS, modified COTS, or with Implementation Partner / Commercial Service Provider / Payload Developer for ISS)
 - [Space Station Research Explorer on NASA.gov](#) (can select 'List All Facilities', then can perform search for specific hardware/facility types)
 - <https://issnationallab.org/research-and-science/partners-and-facilities/implementation-partners/>
- **Within this presentation, NASA is providing the following information to clarify utilization capability needs.**
 - Additional expectations for each phase (Min IOC, CC, FOC)
 - A Tiered Grouping of BLC capabilities to reflect phased groupings of BLC capabilities to enable various scientific investigations,
 - The Offerer can propose providing more capabilities than the minimum requirements for any given Phase (i.e., min IOC, CC, FOC)
 - Offerors can accelerate and minimize incremental phasing of capabilities based on their approach
 - For example, Offerors can directly go to FOC based on vehicle capability as long as timing requirements are met



Tier Groupings – NASA Focus Areas for BLC



Tier Groups	Functional Capabilities
A	<u>Supports (but not limited to) Human Research</u> • Conditioned Stowage (Frozen), Small-Arm Centrifuge, Ultrasound, Atmosphere/Radiation Monitor, Portable Glovebags (in lieu of full glove box), Associated Supplies/Consumables^ <i>NOTE: Tier A required by Initial Operational Capability (IOC), if IOC is being proposed by the Offerer</i>
B	<u>Supports (but not limited to) Life Science & Tech Demos</u>
(B1)	• Additional Conditioned Stowage (Frozen), Workbench, Glovebox, Associated Supplies/Consumables^. <i>NOTE: Tier A + Tier B1 required by Continuous Crew (CC)</i>
(B2)	• Additional Conditioned Stowage (Refrigerated, Incubator), Microscope, Life Science Sample Analysis Capability^, Bio Sequencer^, Cell/Tissue Research Capability^, Analytical Tools and Lab Equipment^, Edge Computing Facility^. <i>NOTE: Tier B2 are additional desired capabilities comprising the Tier B grouping, but not required by CC.</i>
C	<u>Supports (but not limited to) Life Science, Physical Science, Plant Research</u> • Medium-Arm Centrifuge, Additional Microscope, Additional Analytical Tools Lab Equipment^, Plant Research Facility (small), Additional Conditioned Stowage (Frozen) <i>NOTE: Tier A, B and C required by Final Operational Capability (FOC)</i>

Notes

- Items in the same Tier or sub-tier do not have rankings/orderings internal to the Tier
- (^) Details will be provided at draft RFP



Summary Definitions

Minimum Required	Initial Operating Capability (IOC)	Continuous Crew (CC)	Full Operational Capability (FOC)
Number of NASA Crew	2	2	2
Minimum number Total Crew	2	4	4
Minimum Increment Duration	1 month	6 Months	6 Months
Estimated Number of NASA Investigations per Increment	7 - 20 Investigations	20 - 40 Investigations	80 - 100 Investigations
1. Baseline Lab Capability (Provided by CLD) NOTE: Estimated MLE based on NASA hardware experience	10 MLEs volume for suggested Tier Grouping A - Human Research including: Small-Arm Centrifuge, Ultrasound, Atmosphere/Radiation monitors and Conditioned Stowage (Frozen - 1 MLE), Portable Glovebags, Associated Supplies/Consumables	26 MLEs volume for suggested Tier Grouping A Human Research and infrastructure items in Tier Grouping B1 including: Small-Arm Centrifuge, Ultrasound, Atmosphere/Radiation monitors, Glovebox, Workbench, and Conditioned Stowage (Frozen - 3 MLE), Associated Supplies/Consumables	52 MLEs volume for suggested Tier Groupings A-C for Human, Biological and Physical Sciences and to enable Exploration and Technology development in support of human spaceflight goals. This config allows for infrastructure to accommodate the Baseline Laboratory Capabilities list
2. Middeck Locker Equivalent & Deployable Payloads (NASA-provided) Deployable refers to non MLE payloads (ex: aisle mounted, deployed in cabin, etc.) NOTE: Estimated MLE based on NASA hardware experience	2 heritage single middeck lockers or 1 double middeck locker for powered checkout (2 MLE size) 2 Deployable powered payloads	8 heritage single middeck lockers or any combination of single/double quad middeck lockers that equate to 8 heritage single middeck lockers of volume for powered checkout (8 MLE size) 4 Deployable powered payloads	30 reconfigurable MLE volume for single/double/quad sized powered lockers 8 Deployable powered payloads
Estimated Total MLE (BLC & NASA-provided Payloads)	10 + 2 = 12 MLE 2 Deployable	26 + 8 = 34 MLE 4 Deployable	52 + 30 = 82 MLE 8 Deployable
Estimated Cargo Transfer Bag Equivalent (CTBE) Stowage	10 CTBE	26 CTBE	60 CTBE

Note:

1) MLE Volume Estimates are provided based on volumes of existing capabilities on ISS

- 1) A Crew-Tended LEO Destination supporting two (2) NASA Crewmembers for a minimum of one (1) month duration.
- 2) Capability to provide critical functions and systems to safely operate and maintain the on-orbit destination and utilization in LEO, with or without crew onboard, and support Crew and Cargo Visiting Vehicles.
- 3) Provide partial Baseline Laboratory Capabilities to enable (but not limit to) NASA Human Research Utilization:
 - Small-Arm (Lab) Centrifuge, Ultrasound, Atmosphere/Radiation monitors, Portable Glovebags, and Conditioned Stowage (Frozen), Associated Supplies/Consumables
 - Estimated as 10 MLE volume
 - See Tier Grouping A
 - CLD Offerer may propose additional capabilities
- 4) Accommodate 2 heritage single middeck lockers or 1 double middeck locker (NASA provided) for powered checkout & operations purposes (2 MLE) & 2 deployable powered payloads (ex: aisle mounted, deployed in cabin, etc.)
- 5) NASA estimates sponsoring a minimum of ~7-20 investigations
- 6) Provide a minimum of 10 Cargo Transfer Bag Equivalent (CTBE) of soft stowage volume for associated equipment/samples



Continuous Crew Configuration (CC) Definition



- 1) A LEO Destination supporting four (4) total crew members, including two (2) NASA Crewmembers continuously on-orbit with indirect or direct Crew handover for increments of a minimum of six (6) months duration.
- 2) Capability to provide critical functions and systems to safely operate and maintain the on-orbit destination and utilization in LEO, with or without crew onboard, and support Crew and Cargo Visiting Vehicles.
- 3) Provide partial Baseline Laboratory Capabilities to enable (but not limit to) NASA Human Research Utilization:
 - Small-Arm (Lab) Centrifuge, Ultrasound, Atmosphere/Radiation monitors, Portable Glovebags, and Conditioned Stowage (Frozen), Associated Supplies/Consumables, Glovebox, Workbench
 - Estimated as 26 MLE volume
 - See Tier Grouping A & infrastructure items in Tier Grouping B1
 - CLD Offerer may propose additional capabilities
- 4) Accommodate 8 heritage single middeck lockers or any combination of NASA-provided single/double quad middeck lockers that equate to 8 heritage single middeck lockers of volume for powered operations (8 MLE size) & 4 deployable powered payloads (ex: aisle mounted, deployed in cabin, etc.)
- 5) NASA estimates sponsoring a minimum of ~20-40 investigations
- 6) Provide a minimum of 26 Cargo Transfer Bag Equivalent (CTBE) of soft stowage volume for associated equipment/samples



Full Operational Capability (FOC) Definition



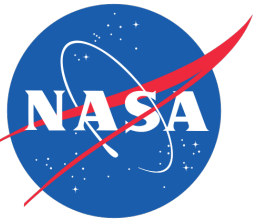
- 1) A LEO Destination supporting four (4) total crew members, including two (2) NASA Crewmembers continuously on-orbit with indirect or direct Crew handover for increments of a minimum of six (6) months duration.
- 2) Capability to provide critical functions and systems to safely operate and maintain the on-orbit destination and utilization in LEO, with or without crew onboard, and support Crew and Cargo Visiting Vehicles.
- 3) Provide the following to enable NASA and Other Government Agency (OGA) efforts including but not limited to Human Research, Biological and Physical Sciences Utilization, and Exploration and Technology development
 - Baseline Laboratory Capabilities as currently defined in CLDP-REQ-1130
 - See Tier Groupings A-C
 - CLD Offerer may propose additional capabilities
 - Estimated as 51 MLE volume
- 4) Accommodate 30 reconfigurable MLE volume for single/double/quad sized powered lockers & 8 deployable powered payloads (ex: aisle mounted, deployed in cabin, etc.)
- 5) NASA estimates this equates to ~80-100 investigations
- 6) Provide a minimum of 60 Cargo Transfer Bag Equivalent (CTBE) of soft stowage volume for associated equipment/samples



Additional Utilization Capability Approach (previously FOC+)



- **NASA is also interested in additional pre-priced services (CLIN 3) that would enhance long-term utilization including but not limited to:**
 - Additional MLE volume for powered payloads
 - Additional deployable payloads (ex: aisle mounted, deployed in cabin, etc.)
 - Additional stowage including ambient and conditioned
 - Additional BLC items to support increased utilization throughput
 - Plus appropriate vehicle resources (i.e. power, data, etc) to support the above items
- **Current forecasts indicate NASA's future needs may be up to:**
 - 48 MLE for powered operations
 - Combination of single, double and/or quad powered lockers
 - 8 deployable payloads
 - 82 CTBE for stowage
- **NASA desires external payload capability but is looking for industry to propose pricing, schedule, and capability (innovation) while still meeting 1130 requirements**
 - This will be priced as optional services under CLIN 3 within the proposal
 - NASA will negotiate at contract award if/when we want to proceed with the services

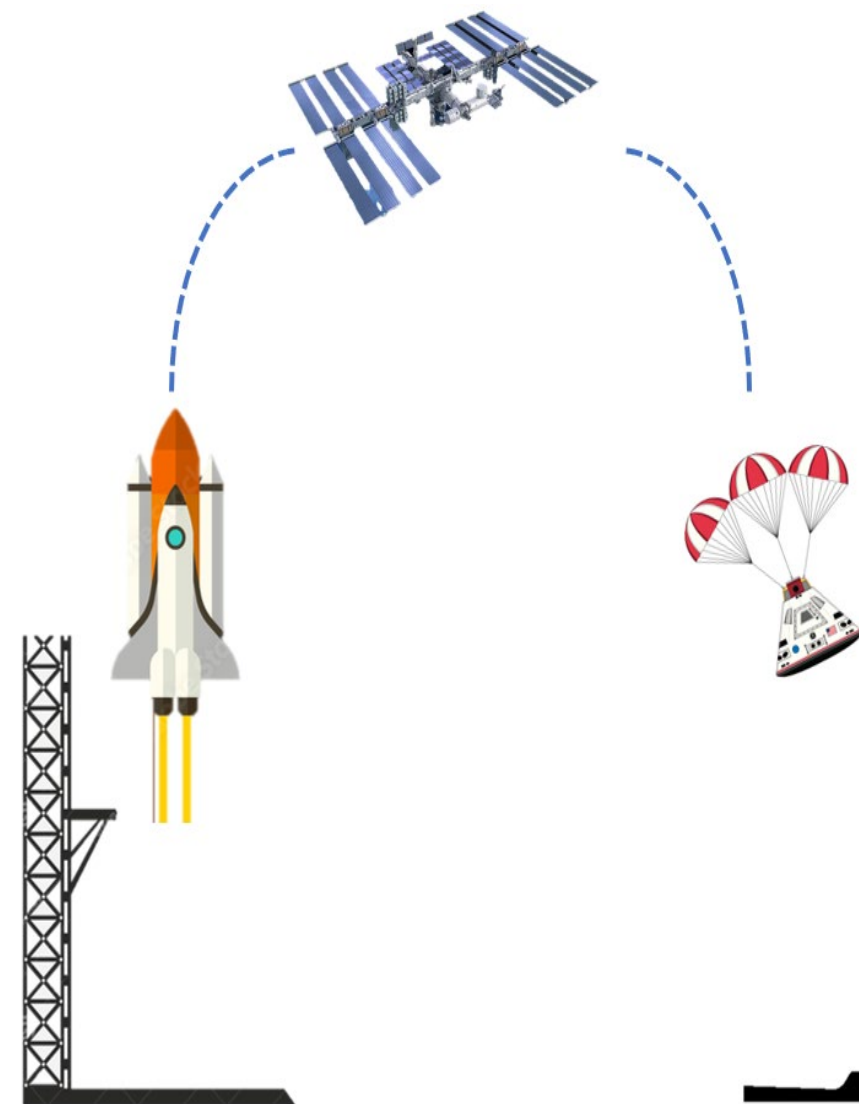


Operations and Training Concepts Overview



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- **NASA has been the leader in human spaceflight operations and training and has developed operational standards governing the processes and best practices based on years of experience dating back to Mercury (1958).**
- **Based on this experience, NASA has developed Operations Standards outlining minimum requirements to ensure Crew safety, Vehicle safety, and Mission success.**
 - The assumption is the provider is responsible for End-to-End services from pre-launch to post-flight recovery including flight and ground operations.
 - The Commercial Provider will need to “meet-the-intent” of the standards provided; allowing the provider to develop and own their processes and use innovative approaches where possible.
 - The following slides provide an overview of the areas of Operations and Training development that need to be covered.





- **The Commercial Provider will be responsible for developing planning products and processes that meet the intent of the standards provided by NASA.**
 - NASA will have internal processes that feed into the Provider's processes for each of these activities and expects to jointly work together to ensure overall safety of the crew and vehicle and mission objectives are met.
- **This may include processes such as:**
 - Trajectory and Mission Design
 - Mission manifests and On-Orbit Logistics planning
 - Procedures and Crew timelines
 - Mission Systems and Training Systems
 - Flight Rules



The Commercial Provider will be responsible for developing and utilizing a process for trajectory design and analysis of nominal and off-nominal trajectories, including dynamic events. This includes the coordination with ground systems (Range, FAA, etc.) to ensure clearance and orbital collision avoidance pre-launch.



The Commercial Provider will be responsible for developing and utilizing a process for mission design to ensure that mission, hardware, software, etc. constraints are identified, as well as the assessments required for integrated operations, dynamic operations, jettisoned hardware, etc.



The Commercial Provider will be responsible for developing a manifest plan for each transportation mission. The Commercial Provider will be responsible for managing how the necessary Destination resources needed for continued operations are acquired, stored, and transported to their final location.



The Commercial Provider will be responsible for developing a logistics management plan. This plan aims to have the right amount of resources, at the right time, that is based on a detailed understanding of resource consumption, necessary maintenance tasks, expected hardware failure intervals, etc.



The Commercial Provider will be responsible for developing and utilizing a process for the generation and maintenance of Flight Rules. Flight rules represent a key component of assigning on-orbit and ground responses for nominal and off-nominal events and are used as a guide for operations personnel. Flight rules act as an operational agreement between the flight operations team and the program management.



The Commercial Provider will be responsible for developing standards for ground and crew procedures that standardizes ground and flight operations procedures as much as possible. This standardization will be an important part of ensure safety of the crew, safety of the ground personnel, and mission success. The Commercial Provider will be responsible for the generation of procedures for both ground and flight for all phases of flight.



The Commercial Provider will be responsible for developing processes and products for the generation, maintenance, and updating of Crew timelines. Crew timelines are used by both ground teams and crew to schedule day-to-day activities and represent a critical function to keep the ground teams and crew in sync. This includes dynamic operations like reboosts, dockings, undockings; and day-to-day tasks like utilization, maintenance, PAO events; in which the ground and crew are required to perform operations together.

- The commercial provider will be responsible for overall mission control of their vehicle and operations.
- Mission Control center for flight operations include, but are not limited to:
 - Mission planning and operations tools
 - Flight Controller personnel
 - Resources for system development and maintenance
 - Data management and storage
 - Building network connectivity to include voice, video, and telemetry data to and from Commercial Provider control center to MCC-H, if required



- The commercial provider will be responsible for ground systems required to support operations and training.
- Facilities and products to support ground systems include, but are not limited to:
 - Launch Pad/Pad support teams
 - Recovery equipment
 - Medical or cargo facilities for post-landing
 - Communications hardware/software to and from Launch Pad/Recovery teams
 - Ground Network
 - Resources for de-integration, refurbishment, and maintenance



- The commercial provider will be responsible for overall training of their ground crews, crew training on vehicle systems including payload provided capabilities, and crew team training.
- Additionally, the commercial provider is responsible for all systems and products required to support the above activities including, but not limited to:
 - Simulators and emulators
 - Mockups for crew/ground training
 - Part Task Trainers
 - Classrooms or Conference rooms for tabletop exercises
 - Products to include lesson plans and supplemental materials used in training





Ground Systems Personnel Training and Certification Approach



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- **The commercial provider will be responsible for the training of ground systems personnel from pre-launch, to ground support of On-Orbit operations, to crew or cargo recovery. This includes:**
 - Launch site support
 - Landing site support
 - Vehicle handling and processing personnel
 - Suit technicians
- **The Commercial Provider will be responsible for identification of safety and mission-critical positions, ensuring that training plans culminate in a certification that is valid for a set period, defines proficiency/currency requirements, and utilizes industry standards to fulfill some certifications, as required.**



Instructor and Flight Controller Training and Certification Approach



- **The Commercial Provider will be responsible for developing instructor and flight controller training and certification process.**
 - An instructor is responsible for class instruction and the certification of other members of the team.
 - A flight controller is responsible for real-time mission support, typically console operations.
- **Instructor and Flight Controller Training can be performed several ways, including, but not limited to:**
 - Simulations with the ability to generate malfunctions
 - Real-time on-the-job training
 - Technical knowledge requirements
 - Classroom demonstrations
 - Training Plan Development
 - Tabletop discussions or exercises



Crew Training and Certification Approach



- **The Commercial Provider will be responsible for developing crew training, scheduling, and integration with external organizations (NASA, other Commercial Partners).**
- **Crew training can be conducted in several different methods including, but not limited to:**
 - Self-study and Classroom lessons with a subject matter expert (SME)
 - High-fidelity multi-system simulations
 - Hands-on hardware
 - Discussions with experienced operations personnel
- **Crew training typically focuses on a specific crewmembers designated responsibilities; however, all crew are trained on a specific set of topics including emergency procedures. At a minimum, flight crew are trained to the level of minimum operational knowledge and skill required to safely operate the integrated space vehicle/element while ensuring mission success.**
- **NASA will provide general astronaut candidate training to NASA crews along with specific training associated with NASA payloads and NASA utilization systems.**



Training for Critical Decision-Making Roles Approach



- **The Commercial Provider(s) will be responsible for generating and certifying all the processes required for Operations and Training execution as the primary authority for CLDS. Including the training of time-critical, risk-informed decision-making roles, including, but not limited to: Mission Management team, flight controller, crew.**



Operations Execution Approach



- **The Commercial Provider will be responsible for developing operations processes that employ a clear definition of authority, roles, responsibilities, and communication protocols as the owner of the Destination.**
 - Flight Control Team responsibilities
 - Operations management
 - Communication and Data Exchange
 - Console Tools
 - Operations Execution
 - Real-time analysis and anomaly resolution
 - Debrief post-flight and lessons learned implementation



Flight Controller Responsibilities



- **The Commercial Provider will be responsible for defining the flight control team (FCT) roles and responsibilities. Based on experience, there are typically multiple flight control teams that support a mission, including, but not limited to:**
 - Transportation Launch Control Team
 - Orbit Flight Control Team, responsible for Destination vehicle (This could be made up of multiple partners and centers)
 - Transportation Landing and Recovery Team



Operations Management



- **The Commercial Provider will be responsible for developing a process for operations management that defines responsibilities for all significant management decisions involving the Commercial Providers options.**
- **This process should establish a clear chain of leadership, including times when decision management is delegated to real-time operators (Flight control team, crew) and when decision require management approval.**
- **NASA expects to participate in these process to ensure decisions are made within the certification baseline and joint flight rules and will have joint go/no go authority over decisions increasing risk to NASA crew.**



Communication and Data Exchange



- **The Commercial Provider will be responsible for developing a process for and any training needed to ensure effective communication and data exchange that includes the full life cycle of the Destination and all associated vehicles. Effective operational communication processes will be critical for ensuring the proper flow of information to all partners and for maintaining situational awareness of the spacecraft among operations personnel and with dependent external Commercial Providers.**
- **This is specifically important when it comes to the voice loop protocols used for communication and the data that will be exchanged between external partners.**
- **The Commercial Provider will be responsible for establishing plans for communication and data exchange between the Commercial Provider, NASA, and other Commercial Partners.**
- **This also includes protection of crew sensitive or medical data.**



- **The Commercial Provider will be responsible for developing tools and processes for the operations team that create a configuration managed documentation documents, requests, and processes used during real-time operations.**
- **Console tools should include capabilities that enable the crew and ground teams to accomplish their missions in a streamlined and effective way. These tools can be developed separately or housed in a single repository and could contain any of the following capabilities.**
 - Anomaly Reporting
 - Real time data exchange of technical or engineering data
 - Uplink requests to crew
 - Crew and ground timeline activities or change request
 - Procedures
 - Points of contact for the flight control team on console
 - Inventory Tracking



- **Executing operations in real time and developing a crew timeline requires multiple iterations and navigating constraints and mission requirements that result in a complex integration effort.**
- **In a “Day-in-the-life” of a mission, the following items need to be considered:**
 - Housekeeping
 - Maintenance
 - Daily Plan Conferences
 - Payload Operations
 - On-Board Training (OBT) or Crew-Based Training (CBT)
 - Exercise
 - Pre-Sleep, Post-Sleep, Sleep
 - Stowage
 - Accessing
 - Tool Gather
 - Meals
 - Video
 - Digital Photography/Earth Photography
 - Adaptation for new crewmembers
 - System Set-up and Tear-Down
 - Hardware Inspections
 - Software/Hardware Reboots
 - Medical Conferences
 - Inflight environment sampling



Real-time Analysis and Anomaly Resolution



- **The Commercial Provider will be responsible for developing a process for real-time analysis during the mission. Following the launch and orbital insertion of a spacecraft, it is possible that the spacecraft does not end up in the orbit design pre-flight. This delta should be analyzed after launch to ensure that mission objectives can still be accomplished.**
- **The Commercial Provider will be responsible for developing a standardized process for the resolution of anomalies inflight. Anomalies are typically discovered by the real-time operational teams and may include any unexpected event, departure from experience or established procedure, hardware/software damage or performance outside of certified design.**



Debrief and Lessons Learned



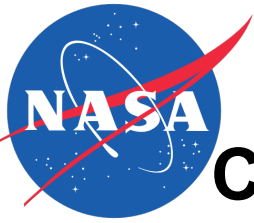
- **The Commercial Provider will be responsible for developing and utilizing a mission debrief and lessons learned process, including the ability to implement, or correct, lessons learned in real time. Mission debriefs represent an important piece of the mission, in which the returned crew can address issues observed during the mission and provides a forum for recommend improvement for future missions.**
- **Crew debriefs may include, but are not limited to:**
 - On-Orbit Maintenance
 - Utilization
 - Training
 - Stowage
 - Habitability



Summary



- **NASA's plan for the commercialization of Low-Earth Orbit is to promote the development and adaptation of innovative solutions that aid in the Commercial Partner(s) developing business model with the NASA operations best practices and requirements as a foundation.**
- **NASA expects strong joint integration and communication in the Operations and Training phases as industry demonstrates their ability to conduct typically government roles and responsibilities.**
 - NASA will evaluate proposed processes, tools and plans based on the Operations Standard to ensure they meet the intent.
- **NASAs requirements and best practices are documented in CLDP-STD-1150 which will be made available no later than draft RFP**



Commercial LEO Destination System Certification Approach



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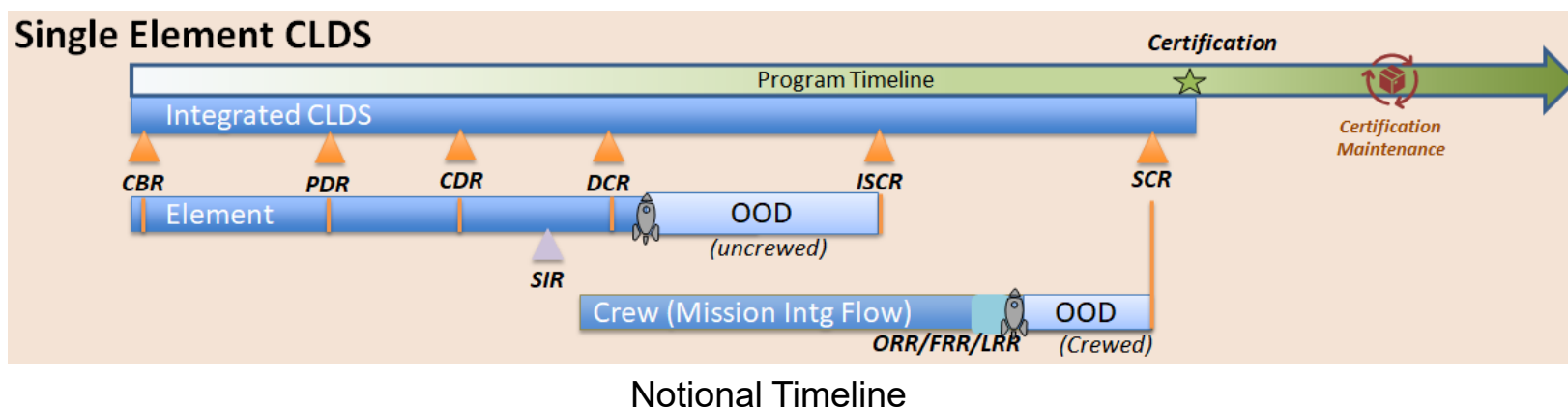


Certification Approach Overview



- **CLDP's Certification approach is dependent upon shared accountability between the commercial provider(s) and NASA for crew safety and mission success and shared assurance across the NASA community to efficiently perform insight and oversight.**
 - This approach is based on CCP and other NASA commercial programs
- **The CLD provider is responsible for DDT&E and retains ownership of their system through operations and eventual retirement**
- **NASA will conduct insight and oversight of the Commercial Provider's CLDS design, development, test & evaluation (DDT&E) and operations efforts.**
- **Certification is intended to address the provider's complete CLDS. This includes the processes, tools and systems required for operations, training, mission integration and utilization. The following areas are included in CLDS Certification:**
 - CLD levied requirements that support the Program's Human Rating Certification (HRC)
 - Focuses on safety of NASA crew and includes functions the Program is responsible for
 - CLDS Functional capability requirements
 - Requirements not related to crew safety that satisfy NASA's NGOs.
 - Crew and Cargo Transportation Certification
- **Once CLDS certification is achieved, NASA and the CLD Provider will assess the CLDS certification baseline against changes identified to processes and/or designs and updates to certification will be completed as required.**
 - Minor updates to the certification may be handled in-line with the M&I/CoFR processes/events.
 - Significant updates (ex. new elements) may require dedicated design and certification milestones
 - NASA will maintain insight and oversight throughout the life of the Program

- The following milestone reviews have been identified to track and finalize certification for CLDs.
 - Contract Baseline Review (CBR)
 - Preliminary Design Review (PDR)
 - Critical Design Review (CDR)
 - Interim Design Review (IDR)
 - System Integration Review (SIR)
 - Design Certification Reviews (DCR)
 - Interim System Certification Review (ISCR)
 - System Certification Review (SCR) (Previously defined as Integrated System Design Review)



Notional Timeline

Definition: Capability – Any element (pressurized or unpressurized), subsystem, or major component that is independently launched and added to the on-orbit destination to meet NASA's requirements for CLDS Certification.



Design and Certification Reviews



- **Certification Baseline Review (CBR) (~ 2 months after Contract Award)**
 - CBR is conducted once at the beginning of the contract to baseline the Provider's Programmatic and Technical maturity
 - Key certification outputs are the CLDS design, Certification Plan, and On-Orbit Demonstration Plan
 - Assess the Commercial Provider's programmatic and technical maturity.
 - Establishes overall Certification strategy and associated applicable milestones required
 - Establish the CLDS architecture, assembly and certification plan
- **Preliminary Design Review (PDR) (schedule to be proposed by Provider)**
 - Demonstrates that the Commercial Provider's preliminary CLDS design (including associated ground and mission systems) meets all system requirements with acceptable risk.
 - Establish the preliminary CLDS design which will be used and referenced at CDR.
- **Critical Design Review (CDR) (schedule to be proposed by Provider)**
 - Demonstrates that the maturity of the Commercial Provider's CLDS design is appropriate to support proceeding with full-scale fabrication, assembly, integration, and test.
 - Define the detailed CLDS design and establish final verification planning prior to execution of the test and evaluation phase of the system and identify the risk to certification.



- **The Interim Design Reviews (IDR) (schedule to be proposed by Provider)**
 - Demonstrate that the maturity of the capability design is appropriate for the design phase, it's impact on the CLDS certification, and the impact on the current on-orbit CLDS and operations.
 - Assess that the technical and programmatic effort is on track to complete the system development, satisfying functional and performance requirements within the identified schedule constraints at an acceptable risk.
 - IDR objectives and EC and SC is a CDR equivalent.
 - The provider may propose multiple IDRs
 - Required for all elements and major components/subsystems launched separately.
 - For single element architectures, this review is not required since scope will be included in System level CDR.
 - The IDR establishes the CLDS configuration which will be used and referenced in the subsequent DCR.
- **System Integration Review (SIR) (schedule to be proposed by Provider)**
 - To assess the readiness of the Commercial Provider to conduct AI&T of the planned capability to be launched.



Design and Certification Reviews



- **Design Certification Review (DCR) (~ L-3 months)**
 - Ensure that the design of the integrated CLDS complies with functional and performance requirements, as demonstrated in verification, validation, and qualification evidence.
 - Ensure that the Commercial Provider has the institutional products, processes, and infrastructure in place to conduct the defined on-orbit demonstration as intended.
 - Establish the CLDS configuration which will be used for the on-orbit demonstration and referenced in the subsequent ISCR.
- **Interim System Certification Review (ISCR) (schedule to be proposed by Provider)**
 - ISCR is derived from the original Integrated System Design Review with a focus on the on-orbit demonstration.
 - Validate that the flown CLDS configuration operates ‘as-designed’ in the intended environment per the planned on-orbit demonstration objectives.
 - Identify any necessary updates to baselined CLDS configuration and operational processes/procedures.
 - Establishes the interim certification for subsequent missions.



Design and Certification Reviews

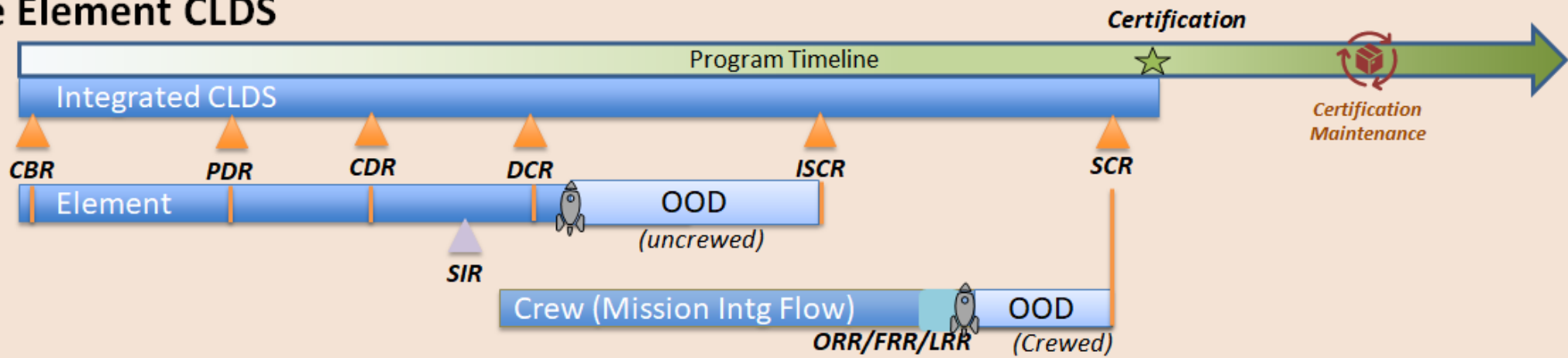


- **System Certification Review (SCR) (schedule to be proposed by Provider)**
 - The purpose of the SCR is to
 - Validate that the integrated CLDS has met all NASA requirements and to address any residual risk to NASA crew safety and mission assurance
 - Establishes the baseline certified CLDS that will be used for NASA services missions.
 - SCR is conducted after the successful completion of all on-orbit demonstrations.
- **On-Orbit Demonstration (schedule to be proposed by Provider)**
 - The purpose of the on-orbit demonstration is to validate the flown end-to-end CLDS in the operational environment. This includes all ground, mission, and space systems required to operate the on-orbit CLD.
- **NOTE: CBR, PDR, CDR, SIR, DCR, ISCR and SCR will be expected regardless of the CLDS design (i.e single element, evolutionary design)**
 - For a single element CLD the above reviews will only need to be conducted once
 - For evolutionary design architectures IDR, SIR, DCR and ISCR will be repeated for each CLDS capability

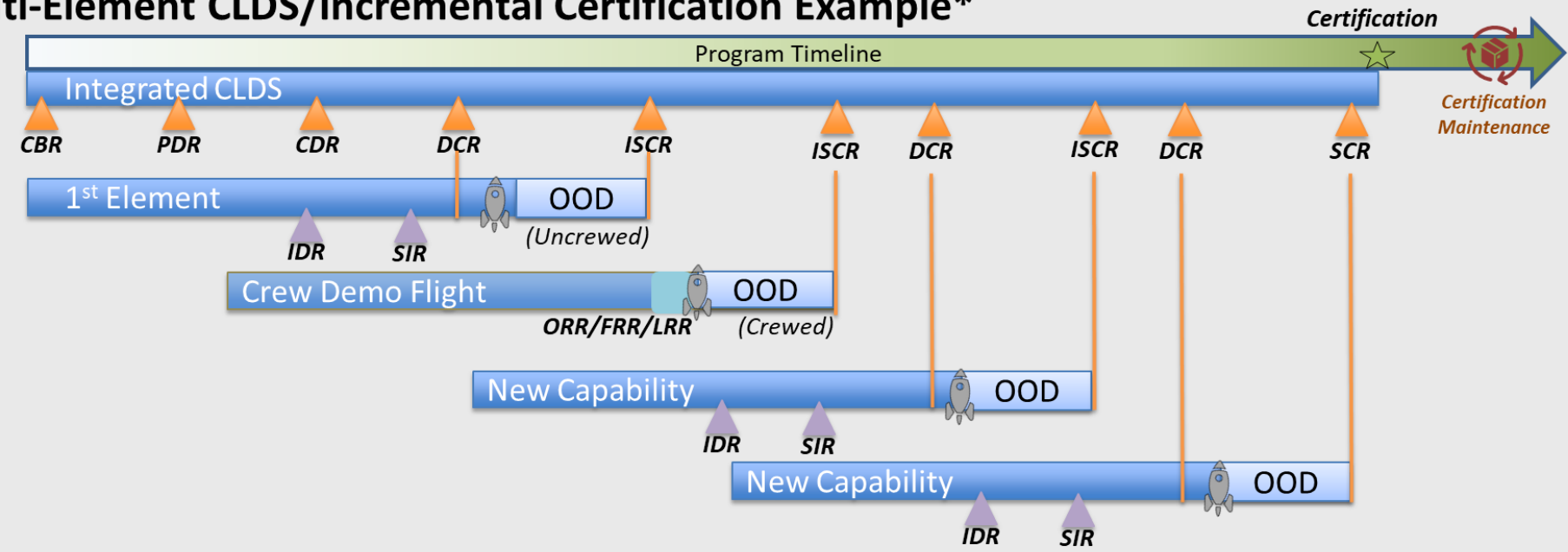


Certification Flow Examples - Notional

Single Element CLDS



Multi-Element CLDS/Incremental Certification Example*



- Integrated CLDS Level Event
- Element/Subsystem Level Event
- Mission Event

*Provider may propose to combine milestones to best support's their development plans. Ex: 1st Element and Integrated CDLS milestones
All information is considered draft and subject to change prior to final RFP release.



Transportation System Certification Approach



- **Certification of transportation vehicles (i.e., Cargo VVs, Crew VV) will be handled in different ways depending on the timeframe, Provider (CLD and VV), and the contracts.**
- **CLD – VV Standard Interfaces**
 - CLD will need to meet standardized VV interfaces to ensure compatibility with Heritage Visiting Vehicles
- **Heritage VVs (Crew and Cargo VVs previously certified by NASA to ISS prior to ISS EOL)**
 - A delta certification will be conducted associated with destination specific interfaces/operations.
 - Forward work on any destination unique mods and schedules will be post contract award
 - Integrated operations (i.e proximity ops and attached ops) is still the responsibility of the Destination provider but will require NASA/Transportation Provider inputs
- **New Transportation services will go through the full certification and be certified to CLDP-REQ-1130 if conducted as part of end-to-end service**



VV Flight Tests and On-Orbit Demonstration Approach

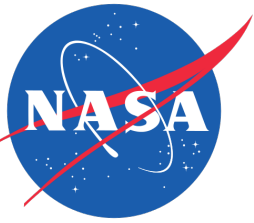


- **VV Flight Tests/On-Orbit Demonstration**

- All VVs will be required to conduct flight tests/demonstrations
- Heritage VVs will conduct a limited flight test prior to docking based on contract negotiation.
 - Flight tests/demonstrations are expected for delta certification areas including integrated operations (i.e Prox Ops and attached ops).
 - May occur as part of first mission to the CLD (Crewed or uncrewed)
- New vehicles may require both an uncrewed and crewed demonstration flight based on follow on contract negotiation

- **CLD On-Orbit Demonstration**

- CLD Providers will be expected to conduct uncrewed and crewed on-orbit demonstrations
 - An Uncrewed demonstration/check out will be expected prior to on-orbit crewed demonstration and checkouts.
- The on-orbit demonstration objectives and criteria will be detailed for NASA's review in the Commercial Provider's On-Orbit Demonstration plan in conjunction with their Certification Plan.



CLDP Oversight and Insight Approach



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Introduction – Insight and Oversight Definition (NFS 1846.401)



Per the **NFS 1846.401**, *Government Contract Quality Assurance*, part (a) the Government may conduct inspection and other quality requirements through insight or oversight as defined in (i) and (ii) below (emphasis added):

(i) “**Insight** requires the **monitoring** of contractor quality data and Government identified metrics and contracted milestones and may also involve the review of contractor work procedures and records. Insight is a continuum that can range from low intensity, such as reviewing quarterly reports, to high intensity, such as performing surveys and reviews.”

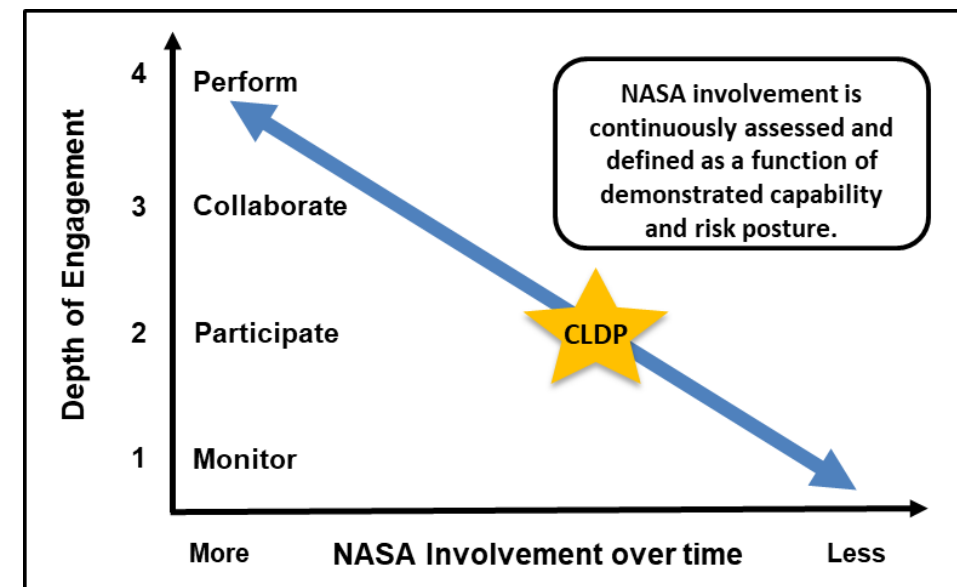
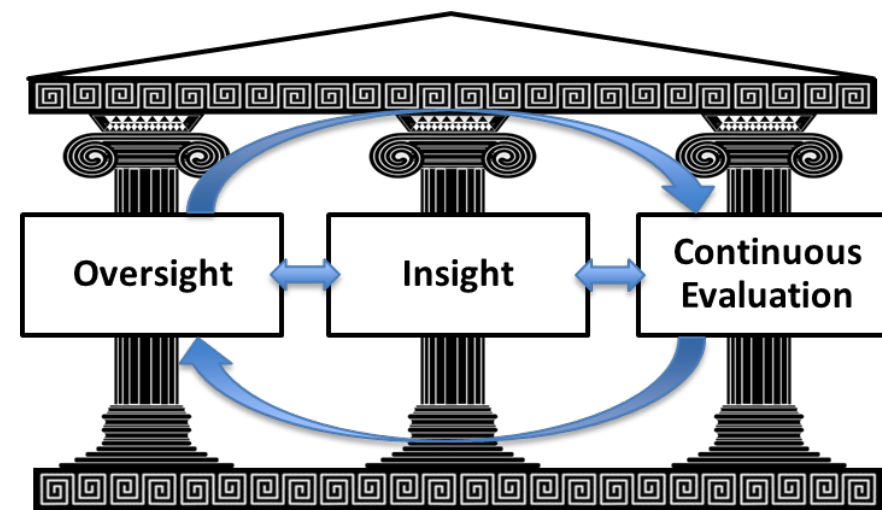
The government will be given access to “Insight” information/data as described by the contract via PWS and/or Section H.25 Insight clause. Access is extended to Support Service Contractors.

(ii) “**Oversight** occurs in line with the contractor's processes. **The Government retains and exercises the right to concur or non-concur with the contractor's decisions** affecting product conformity. Non-concurrence must be resolved before the contractor can proceed. Oversight is a continuum that can range from low intensity, such as quality system audits, or Government concurrence in reviews (e.g., preliminary design review or critical design review), to high intensity oversight in which the Government has day-to-day involvement in the contractor's decision-making process (e.g., mandatory hardware inspections or process witnessing). The decision to use insight or oversight is based on an assessment of the risk inherent in the activity being surveilled. Surveillance must be conducted whether or not the contract effort has been structured as a performance-based acquisition.”

Government Oversight provides NASA CLDP authority and/or approval over designated Contractor activities, processes, documents, plans, requirements, reviews, milestones, deliverables, safety, schedule, and data in support of the Contractor's DDT&E, Missions, and NASA's use of the CLDS end-to-end services.

source: <https://prod.nais.nasa.gov/far/far0595-nfs012617/1846.htm>

- For CLDs, NASA is implementing an approach consisting of three pillars:
 - **Oversight:** The set of Commercial Provider activities that NASA has approval authority over
 - **Insight:** The set of Commercial Provider activities that NASA will monitor or participate in to gain an understanding of the Contractor's activities and data through watchful observations, inspections and interactions, without approval or disapproval authority
 - **Continuous Evaluation:** The risk-based process that continually assesses all Provider information to verify or improve the Oversight Activities or adjust the selected Insight DoE Levels as needed.
- Objective is to lessen NASA involvement over time





- **Oversight activities are those that NASA has approval authority over**
 - Focused on completion of CLDS Certification, Verification Closure Notices and acceptance of performance milestones for core work and mission task orders
- **Oversight activities include:**
 - Program Milestone Reviews
 - Program Readiness Reviews
 - Contract Mandated Checkpoints (e.g. real-time polls)
 - Type 1/2 DRDs (Hazard Reports, certification packages, V&V Plan)
 - Verification Closure Notices (VCNs)
 - Trusted Vendor Authorization
 - CLDS Certification and Demonstration
 - Integrated operations with the NASA Crew, NASA ground and NASA Flight Hardware
 - NASA Led Mishap and Failure Review Boards
 - Crew Selection for NASA Missions
 - Test Readiness Reviews involving NASA personnel

- **Insight activities are opportunities for NASA to monitor partner/provider (and potentially subcontractor) performance in between oversight product deliveries. Insight is enabled through the contract. (Insight DRD, Right of access clause, etc.)**
 - Used to gain an understanding of the Contractor's activities and data through watchful observations, inspections and interactions, without approval or disapproval authority
- **Planned insight activities include:**
 - Provider/Partner Meeting Participation
 - Planned Surveillance and Product Assurance Actions (PAAs)
 - Joint NASA/Partner participation in activities
 - Quality Data Review / Data Reviews / Desk Audits (enabled as part of contract insight DRD)
 - Audits - Used for: High Level oversight, full process review for areas with quality concern
 - Walkdowns / Process Reviews - Used for: High Level insight, full process review for areas with little/no current visibility
 - Targeted daily observation/Resident Office awareness - Used for: Low level insight, full process insight for potential quality concerns, but currently have little/no current visibility. (Reoccurring)



- **CLDP is planning to use a Risk-Based approach to determine the level of government engagement required for oversight and insight activities**
 - Similar to process utilized by CCP to assess Hazard controls to define highly-critical hardware that required government manufacturing surveillance
 - Two critical components:
 - Assessment of confidence in critical provider processes
 - Assessment of risk in specific design choices
- **CLDP's risk-based assessment framework will help the government focus on areas with significant unknowns or perceived risk**
 - Risk-based framework will be used to evaluate new data, events, circumstances etc throughout program execution to inform CLDP decision making
 - CLDP can use assessments of provider processes and risks to determine whether increasing or decreasing NASA's level of engagement is appropriate



Trusted Vendor Approach



- **Trusted Vendor process transitions tasks/work traditionally done by NASA to a Partner/Provider (ISS Safety refers to it as franchising)**
 - Currently used with IPs, commercial providers, and on some PAM equipment
- **Trusted Vendor framework includes 2 parts.**
 - Authorization
 - NASA reviews their processes and assesses their performance and authorizes them when they have demonstrated that they are ready
 - Maintenance
 - Insight activities used to ensure that performance is within acceptable limits
 - Insight activities are established when Trusted Vendor Authorization is granted
 - Offline document reviews
 - Process “ride alongs”
 - Escapes/successes/feedback on outcomes
 - Maybe adjusted based on performance
- **NASA envisions using a Trusted Vendor process to decrease NASA oversight over time in the areas of Safety, Operations and Utilization as the commercial provider demonstrates over time consistent capability in these areas**
 - NASA working to develop criteria for transition from more jointly managed work to CLD full delegated activities

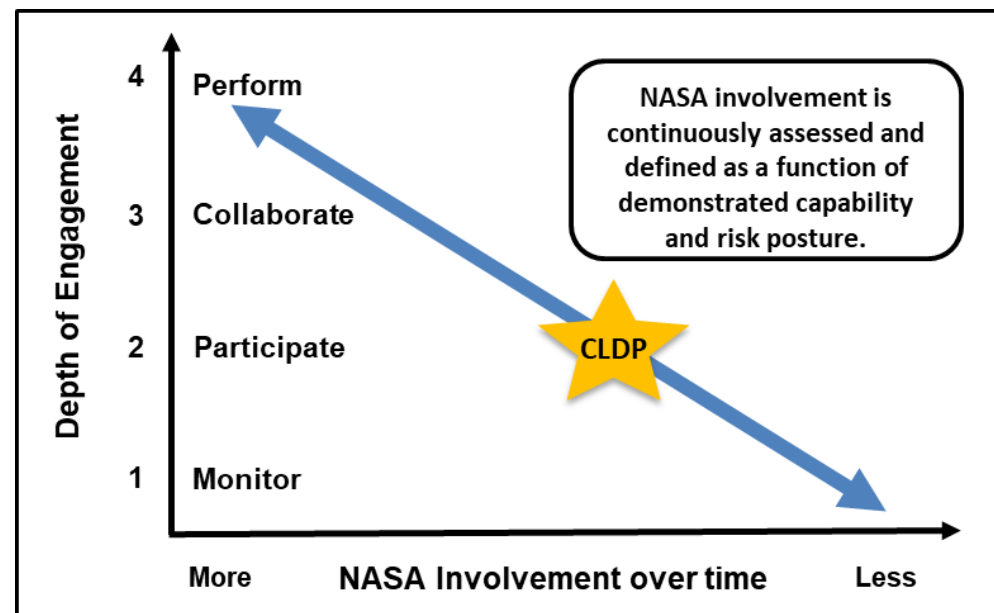


Risk Based Assessment (RBA) Approach



- **NASA is planning to use a Risk-Based approach to determine the level of government engagement required for oversight and insight activities**
 - Similar to process utilized by CCP to assess Hazard controls to define highly-critical hardware that required government manufacturing surveillance
 - Two critical components:
 - Assessment of confidence in critical provider processes
 - Assessment of risk in specific design choices
- **CLDP's RBA process will help the government focus on areas with significant unknowns or perceived risk**
 - NASA can use assessments of provider processes and risks to determine whether increasing or decreasing NASA's level of engagement is appropriate (next slide)

- **Level 1 – Monitor Commercial Provider CLDS design, development, and operational activities like DRD Approval/Reviews, Milestone Reviews, program management reviews, Operations Activities, etc**
- **Level 2 – Participate in reviews and Technical Interchange Meetings, review lower-level data and supplemental documentation, performing periodic audits on pre-defined processes**
 - All data used to support the development of DRDs, ex., review Test Plans and Procedures, observe Test execution, etc.
- **Level 3 – Collaborate in routine Working Groups and Technical Meetings to identify and resolve issues, and a evaluate task details**
 - Participate in the development of lower-level activities, ex., participate in the development of Test Plans and Procedures,
- **Level 4 – Perform independent modeling and analysis**

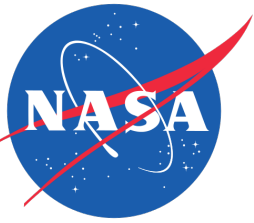




Summary and Forward Work



- **NASA is developing CLDP-PLN-3016 CLDP Oversight and Insight Management Plan to outline the overall expectations for NASA Insight and Oversight.**
- **The Commercial provider will deliver their Insight Implementation Plan outlining their approach for providing NASA the required Insight and Oversight needed as outlined in the contract and the above plan.**
- **This document will be delivered NLT draft RFP for industry comment.**



Mission Integration Approach



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Mission Integration Approach Introduction



- **The Mission Integration process ensures the proper identification, tracking, handling, and communication of programmatic needs to execute the CLDP Mission**
- **The Mission Integration function consists of two main functions**
 - Strategic or multi-increment planning
 - Tactical planning; pre-flight, during the mission or increment and post-flight reporting
- **The Commercial CLD Providers are responsible for the development and implementation of their own mission integration plans and NASA will partner plans for successful execution**
- **The Mission Integration function facilitates the recurring Programmatic Reviews during the tactical phase to ensure the NASA and CLD Providers progress are on track for each mission**



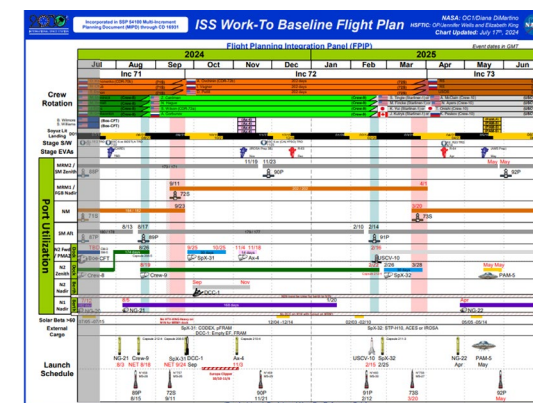
Mission Requirements Approach



- **Task Orders will ‘order’ a mission providing the contractual commitment from NASA for the specific targeted mission dates, utilization requirements and mission objectives, ATP criteria, crew and cargo complement, initial mission success criteria, and mission milestone review payment schedules**
 - Estimated Task Order initiation at approximately launch minus 24 months
 - The provider is required to meet the ATP criteria before NASA will grant ATP for a particular mission
- **Mission specific data is managed via Data Requirement Deliverables (DRDs) outlining the tasks, data delivery requirements to and from NASA, and the associated timelines for integration and execution of a NASA mission**
- **It is expected there will be CLD Provider unique timelines, responsibilities and handoff points depending on provider’s design and mission integration approach**

- **Strategic or multi-increment planning provides the long-range planning for NASAs crew and utilization needs**
 - This includes definition of the Flight Plan, Increment, resources and accommodations available at the Increment level as assessed against transportation and crew rotation requirements
- **The planning process begins by performing an integrated traffic planning analysis to define the transportation support available for delivery of crew, cargo, utilization, and consumable resupply to the CLD**
 - The CLD provider is responsible for the definition of assembly and outfitting hardware, spares, crew supplies, maintenance, logistics, and consumables resupply to the CLDs
 - It is envisioned that NASA will participate in this process via joint working groups
- **The result of this iterative process is a CLD Flight Plan**
 - This is performed by integrating the CLD's transportation system capabilities and each crew and vehicle support cargo transportation requirements into a plan that satisfies the utilization and increment objectives
 - The plan results from balancing the vehicle traffic plan, crew rotation plan, cargo manifest plan, and crew and cargo vehicle capabilities
- **During the GFF phase, and especially during the overlap with ISS, this is envisioned to be a bilateral process to develop the integrated traffic plan**

Example of a complex ISS Flight Plan





Tactical Planning Approach



- **The tactical planning process begins approximately 24 months prior to the start of an Increment with the initiation of a task order and ends with post-flight/post-increment assessments, reports and lessons learned**
 - Tactical planning and manifesting will be performed for each Increment by the CLD Provider with inputs from NASA
- **Tactical planning and manifesting defines the resources, allocations, research objectives, priorities, and manifests for each flight and increment**
 - The CLD provider will be responsible for integrating all customer and destination needs into the integrated manifest
 - The CLD Provider will be responsible to define the requirements and hardware for outfitting, spares, logistics, maintenance, consumables, crew supplies and crew from other customers
 - NASA is responsible for collecting, integrating, and tracking all NASA increment utilization requirements, utilization manifests, and priorities through the end-to-end mission
 - During the time period of NASA providing GFF, roles and responsibilities will have to be jointly developed, coordinated and managed with NASA and the CLD provider
 - NASA is still assessing the option of providing manifest integration and cargo packing services during the GFF period.



Certification of Flight Readiness (CoFR) Strategy Approach



- **Certification of Flight Readiness (CoFR) is the attestation that the joint NASA and CLDS Providers mission plan satisfies all vehicle and mission specific requirements, and all parties are ready to safely conduct the mission**
 - This certification includes pre-launch activities, launch, dock/berth, on-orbit operations, undock/unberth and return/ disposal and its accompanying cargo readiness, manifested launch and return cargo readiness; on-orbit configuration readiness; and ground facilities, ground support personnel, flight controllers and flight crew readiness to support defined events.
- **The CLD Provider is accountable & responsible for the DDT&E of the CLDS and the end-to-end mission execution**
 - During the GFF transportation phase, NASA and the GFF contractor will be responsible for the CoFR of the transportation services
- **NASA is accountable for ensuring compliance to NASAs human spaceflight requirements thru evaluation and approval of the Contractor's compliance evidence and execution of NASA's insight into the Contractor's solution in accordance with a risk-based insight approach implemented under a shared assurance model**
- **NASA is also accountable & responsible for the certification of NASA payloads to ensure that the payloads are safe to launch and operate as evident by CLDP safety and interface verification products including software; and operational product and personnel readiness.**
 - Functional performance and mission success of payloads is the responsibility of the payload developer
 - NASA will also be responsible for the CoFR of GFS services negotiated and defined in the CLDC contract
- **Contractual PWS/DRD requirements will define the minimum NASA CoFR requirements, certification and mission integration milestone reviews with entrance and success criteria, and identify CLD Provider participation and deliverables to be provided**
- **The CoFR process culminates in an FRR that will in the end conduct a Go/No Go poll to proceed with the mission**



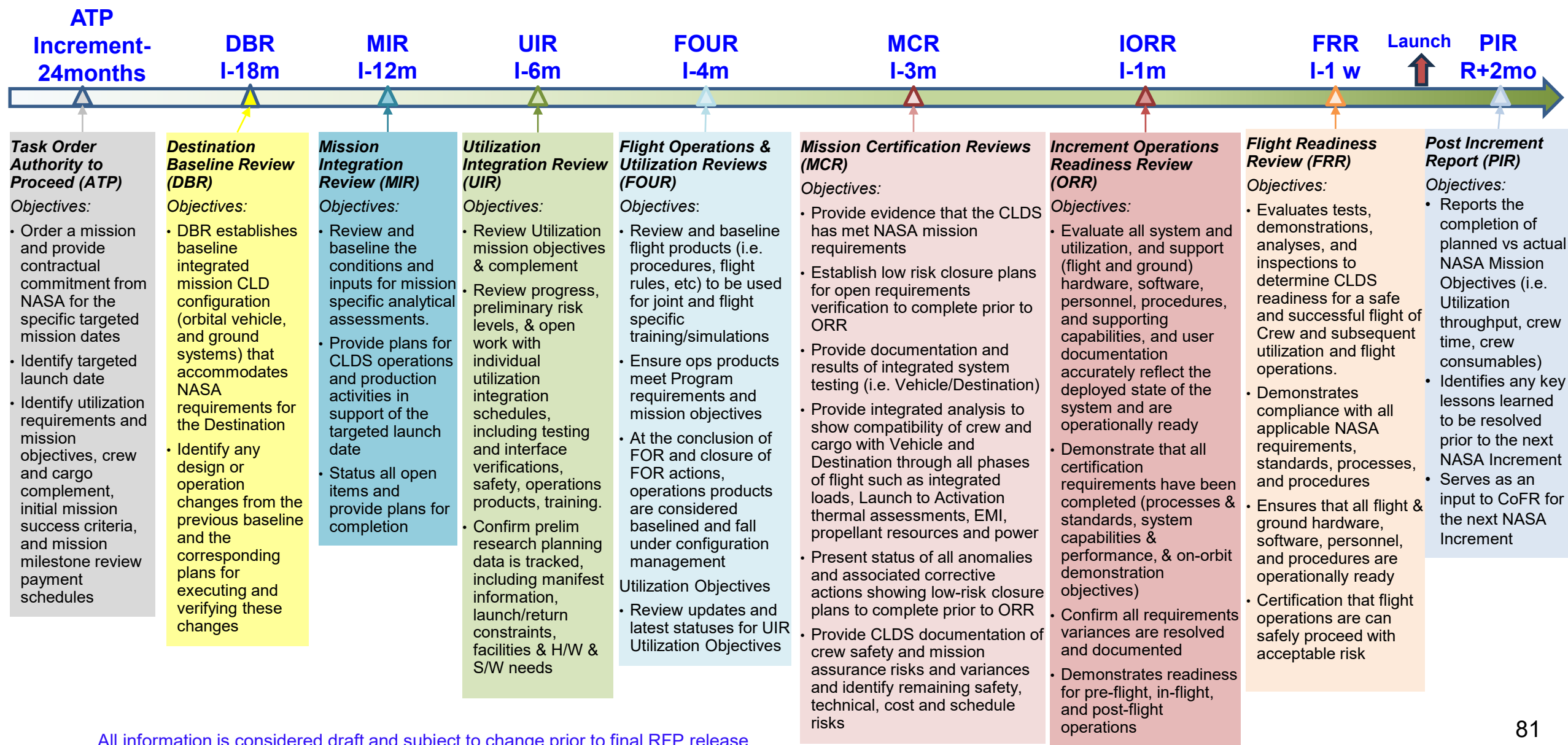
Mission Integration Template Proposal



- **NASA has developed template CLD Mission Integration milestones as described in the subsequent charts**
 - The template defines proposed schedules, milestone objectives, and product deliverables from Mission Authority to Proceed (ATP) through launch
 - Transportation Integration (Cargo and Crew) and Certification milestones are defined separately
- **The CLDS Mission Integration template and associated deliverables are designed to:**
 - Understand changes from the last baseline and establish the on-orbit mission configuration
 - Gain insight into the CLD Providers mission readiness to support all NASA Mission Objectives
 - Gain insight into the evolution of the CLD Providers strategic plan to a detailed tactical plan
 - Refine launch opportunities from a broad window to a viable target launch date
 - Confirm incorporation of all NASA requirements and priorities into the baseline Increment mission design
 - Mitigate or accept all known risks
 - Understand and disposition all identified anomalies and issues
 - Receive iterative Contractor product deliveries at regular milestone intervals
 - Provide Certification of Flight Readiness (CoFR) endorsements for a specific mission
- **NASA is seeking industry feedback on this Mission Integration template to achieve maximum efficiency while still meeting all NASA defined mission objectives and priorities**



Proposed Mission Integration Milestone Review Template





Proposed Destination Baseline Review



- **Destination Baseline Review (DBR)**

- Contractor-Led milestone with NASA participation
- No later than 18 months prior to the Increment (I-18 mo.)
- Establishes the integrated mission CLDS configuration (on-orbit platform, visiting vehicles, and ground systems)
- Identify all design, configuration, or operational changes from the previous certified baseline.
- Present plans to incorporate and certify changes to the design baseline.
- Define all CLD reconfiguration, installations, or assembly activities occurring during the targeted Increment while NASA crew is on-board
- Identify all the Commercial Activities the CLD Provider is planning for the Increment
 - Including use of NASA crew or NASA hardware



Proposed Mission Integration Review



- **Mission Integration Review (MIR)**

- Contractor-Led milestone with NASA participation and inputs
- No later than (NLT) twelve 12 months prior to the increment (I-12 mo.)
- Confirm the targeted launch window and identify any known risks that could disrupt launch availability
- Define the combined Contractor and NASA Crew compliment for the Increment
- Baseline the NASA Crew, CLD Crew, cargo, and utilization requirements for the end-to-end mission
- Confirm the initial conditions and planned data deliveries for the Contractor's mission specific analysis with a plan and schedule for completion in support of the targeted launch date
- Establish the plan and schedule for operational product development and production activities in support of the targeted launch date
- Identify all the Commercial Activities the CLD Provider is planning for the Increment
 - Including use of NASA crew or NASA hardware
- Review the status of all open items and closure plans identified at the DBR



Proposed Utilization Integration Review



- **Utilization Integration Review (UIR)**

- Contractor-Led milestone with NASA participation and inputs
 - Includes NASA products, CLD Provider products, and Joint products development status
- No later than (NLT) six (6) months prior to the Increment (I-6 mo.)
- Review the utilization mission objectives and mission manifest, including science and research and technology demonstration payloads
- Review progress, integration risk levels, and open work with individual utilization integration schedules, including testing and interface verifications, safety, operations products, training
- Review integrated/complement level data such as facility throughput models, planning information, operations or spatial or other conflicts, and assess risk to successful operations
- Review preliminary research planning data is being tracked, including manifest information, launch/return constraints, facilities needs, and other hardware/software needs



Proposed Flight Operations and Utilization Review



- **Flight Operations and Utilization Review (FOUR)**

- Contractor-Led milestone with NASA participation and inputs
 - Includes NASA products, CLD Provider products, and Joint products development status
- No later than (NLT) four 4 months prior to the Increment (I-4 mo.)
- Review and baseline flight products (i.e. procedures, flight rules, etc) to be used for joint and flight specific training/simulations to ensure ops products meet Program requirements and mission objectives.
- Review the utilization mission objectives and mission manifest
- Review the Contractor's progress, integration risk levels, and open work to satisfy the individual NASA utilization integration schedules
- Define all completed or planned payload interface tests, verifications, and safety reviews
- Provide a status of payloads operational product development and crew utilization training
- Characterize the integrated CLD utilization facility throughput models, strategic plans, conflicts, and risks to complete all utilization mission objectives
- Confirm that all defined research constraints are tracked and are accommodated including ground, launch, and return handling constraints, late-load, early-access, and early de-stow access requirements,
- Define any on-orbit payload facility hardware and software needs (including upgrade plans)
- Baseline the integrated utilization priority list for the Mission
- Review the status of all open items and closure plans identified at the MIR.



Proposed Mission Certification Review



- **Mission Certification Review (MCR)**

- Contractor-Led milestone with NASA participation and inputs
 - Includes NASA products, CLD Provider products, and Joint products development status
- No later than (NLT) three (3) months prior to the Increment (I-3 mo.)
- Provide evidence that the CLDS has met all NASA requirements (Phase 3 HRs & VCNs approved) or has a viable plan for closure prior to the ORR.
- Review the status and schedule of all remaining standard and non-standard open work and confirm the Contractor has a viable plan for completion or risk acceptance prior to the ORR.
- Confirm the crew and cargo handover schedules and any required special accommodations are or will be in-place at the time of handover
- Provide documentation and results of the Contractor's CLDS integrated system testing and analysis in comparison to the planned on-orbit configuration for the mission (including compatibility of crew and cargo vehicles)
- Present status of all anomalies and associated corrective actions showing low-risk closure plans prior to ORR
- Present crew safety and mission assurance risks and variances and identify remaining safety, technical, cost and schedule risks
- Confirm all CLDS infrastructure, consumables, facilities, personnel and services are in place and will be ready for the crewed mission
- Confirm the targeted launch date within the defined launch window
- Identify all the Commercial Activities the CLD Provider is planning for the Increment
 - Including use of NASA crew or NASA hardware
- Review the status of all open items and closure plans identified at previous reviews



Proposed Increment Operations Readiness Review



- **Increment Operations Readiness Review (IORR)**

- Proposed to be Jointly-Led Co-chaired Milestone
 - Includes NASA products, CLD Provider products, and Joint products development status
- No later than (NLT) one (1) month prior to the Increment (I-1 mo.)
- Ensure the integrated CLDS support hardware, software, personnel, procedures, and documentation are in-place and accurately reflect the NASA mission objectives for the current state of the CLDS.
- Demonstrate readiness for pre-flight, in-flight, and post-flight operations
- Confirm all applicable certification and mission integration requirements have been satisfied or have a viable plan for closure
- Confirm all requirements variances are resolved and properly documented
- Confirm all relevant schedule, technical, and programmatic risks are identified and appropriately dispositioned
- Present status of all anomalies and associated corrective actions showing low-risk closure plans prior to FRR
- Identify all potential CoFR Exceptions
- Identify all the Commercial Activities the CLD Provider is planning for the Increment
 - Including use of NASA crew or NASA hardware
- Conduct a Go/NoGo-Poll to confirm readiness to proceed to the Flight Readiness Review (FRR)



Proposed Flight Readiness Review



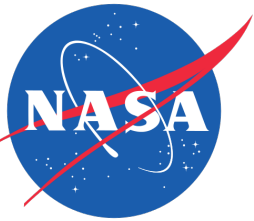
- **Flight Readiness Review (FRR)**

- Proposed to be Jointly-Led Co-chaired Milestone
 - Includes NASA products, CLD Provider products, and Joint products development status
- No later than (NLT) five days prior to the Increment (I-5 days)
- Confirm the integrated CLDS readiness for a safe and successful launch and to execute the end-to-end mission
- Ensures all flight and ground hardware, software, personnel, and procedures are operationally ready for pre-flight, increment, and post-flight operations
- Review tests, demonstrations, analyses, and audits attesting the system's readiness for a safe and successful end-to-end mission.
- Confirm all relevant schedule, technical, and programmatic risks are identified and appropriately dispositioned
- Confirm all requirements variances are resolved and properly documented
- Confirm compliance with all applicable requirements, standards, processes, and procedures
- Ensure all flight & ground hardware, software, personnel, and procedures are operationally ready
- Document all CoFR exceptions and confirm all CoFR exceptions have been formally dispositioned
- Conduct a Go/NoGo Poll to proceed to the Launch Readiness Review



- **Post Increment Report (PIR)**

- Preliminary report delivered at no-later-than Increment Crew return + two weeks (R+2w)
- Final report at no-later-than Increment Crew return + two months (R+2m)
- Reports the completion of planned vs actual NASA Mission Objectives (i.e. Utilization throughput, crew time, crew consumables)
- Identifies any key lessons learned to be resolved prior to the next NASA Increment
- Serves as an input to CoFR for the next NASA Increment



Government Furnished Services Approach



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Government Provided Services



- **In order to facilitate a timely transition from ISS to CLDs, NASA plans to offer government provided services in the following three categories:**
 - General Offered services (GTA/RSAAs)
 - Government provided Transportation and associated services
 - Additional areas of collaboration



General Offered Services (GTA / RSAA)



- **NASA plans to offer services to industry to support development of future Commercial LEO Destinations utilizing GTA/RSAA's**

1) Pursue a “GTA” – Government Task Agreement

A GTA is a formal agreement between the government performing organization and the partner that contractually commits them to deliver agreed upon goods/services at cost (addressed via contract)

Upon partner request, government performing organization will provide a SOW & quote for work to be performed to the partner. If agreed upon, the partner incorporates the government work package into their proposal. If the partner gets selected, NASA work packages will be executed

The GTA mechanism can only be used by partners that have an existing funded NASA contract and require NASA Program approval. Scope is limited to the NASA Program of Record only – tied to CLINs*

2) Pursue a “RSAA” – Reimbursable Space Act Agreement

A RSAA is a formal agreement between the government performing organization and the partner that contractually commits them to be reimbursed for agreed upon goods/services

Performing orgs will provide a SOW & quote for work to be performed to the partner. If agreed upon, the agreement is enacted and NASA work packages are executed

RSAA's scope can encompass any partner program or effort

RSAA's require NASA Center and HQ approvals

- **Information related to general services at each NASA center is provided with this presentation and will be posted under the Technical Library on sam.gov. A final list of services will be provided with the Draft RFP.**

If interested, please reach out to the NASA Center POC. They will guide you through the process & coordinate work packages/quotes with performing orgs



Schedule and CLDP NASA POCs (GTA/RSAA)



- Offerors begin discussions with Center POCs to determine services desired from the government
- Offerors proposed draft agreements due to NASA Center Partnership Office POC(s): NLT 15 days after Final RFP release
- NASA Center approved agreements provided to Offeror: ~30 days later
- Offerors choose which ones to submit: Provided with Proposal

CLDP GTA POC: Michael Berdich, michael.a.berdich@nasa.gov

Center	Name	Phone	Email
Ames Research Center	Sharmilla Bhattacharya	650-604-1531	sharmilla.bhattacharya@nasa.gov
Glenn Research Center	Bob Corban	216-433-6642	robert.r.corban@nasa.gov
Goddard Space Flight Center	Mark Lupisella	301-286-2918	mark.l.lupisella@nasa.gov
Jet Propulsion Laboratory	David Eisenman	818-667-8737	david.j.eisenman@nasa.gov
Johnson Space Center	Steven Gerker		steven.f.gerker@nasa.gov
Kennedy Space Center	Erik Holbert	321-289-5038	Eirik.holbert-1@nasa.gov
Langley Space Center	Chris Giersch	757-864-6590	christopher.e.giersch@nasa.gov
Marshall Space Flight Center	Justin Jackson	256-701-7204	justin.r.jackson@nasa.gov
Stennis Space Center	Jason Richard	228-363-4810	jason.a.richard@nasa.gov

LESSONS LEARNED

GTA process is dynamic and requires a large amount of coordination and back forth conversations to iterate and negotiate.
Encourage early GTA coordination to provide Offeror time to iterate with NASA Center performing organizations



Government Provided Transportation Approach



- **NASA is offering transportation, from the heritage vehicle fleet, for initial NASA CLD missions (Crew and Cargo)**
 - NASA will provide transportation for an initial set of missions, by contracting directly with transportation providers, and transition to CLD end-to-end service (including transportation)
 - NASA Missions are required to utilize NASA provided transportation until end to end service transition
- **NASA plans to order full crew and cargo missions from Transportation providers and plans to offer excess capacity/seats to the provider**
 - NASA envisions mixed missions including NASA, IP and other private astronauts
 - Cost reimbursements to the Gov't for use of transportation for non-NASA customers is expected
 - CLD Providers may propose an alternative approach as part of their proposal if they do not want to utilize NASA procured transportation
 - CLD Provider is responsible for Private Astronaut Missions, assembly or cargo outfitting missions
 - Provider can propose that NASA provides or partners with Provider to provide cargo transportation for outfitting flights, if those outfitting flights include NASA needs
 - NASA may choose to on-ramp new certified vehicles and/or utilize International Partner provided vehicles for transportation missions



NASA Provided Crew and Cargo Transportation Approach



- **NASA is considering to have the offerors provide as part of the proposal the number and type of flights to be provided by NASA and the transition plan**
 - Table 1 provides an example of the phasing and quantities of NASA provided transportation along with the transition to Commercial Provider end to end service.

Notional Table

Table 1 CLDP MISSIONS (E.G. FLIGHTS) PER FISCAL YEAR							
	FY29	FY30	FY31	FY32	FY33	FY34	FY35
Crew (NASA Provided as GFF)		2	2	2	1		
Crew (CLD Contractor Provided)					1	2	2
Cargo (NASA Provided as GFF)		2	3	3	2		
Cargo (CLD Contractor Provided)				1	2	4	4

- Mission specific task order pricing should account for NASA provided transportation
- Final flight quantities and types will be negotiated as part of the source selection process based on NASA's budget

Details of how NASA is going to handle GFF Transportation is still under development



Government Furnished Transportation Services



The following services are being proposed to be provided by NASA for all Government Furnished Flights:

Service

Emergency Response at Pad (e.g. Pad rescue, Triage sites, Medivac, DMCF Readiness)

Transportation to Launch Site for NASA Crew and Family and NASA HSP Personnel

Crew Rescue (e.g. Big Picture, Pad Aborts, Ascent, World Wide Free Flight & Landing)

Transportation back to Houston for NASA Crew (Nominally and from Designated Medical Facility if required)

Baseline Data Collection (BDC) for NASA Crew: NASA (Conducts BDC)/Provider (Coordinates Facility availability)

The following services are still being evaluated to be provided by the government as part of Government Furnished Flights, may be joint or may be the responsibility of the Destination provider:

Service

Visiting Vehicle Manifest Integration (All visiting vehicles)

Cargo Bag Packing (All visiting vehicles)

Cargo Safety Assessment and Integrated Bag Level Hazard Analysis (IBLHA)

Cold Stowage equipment, facility, and integration support (e.g. Packing, Transport to/from VV and facility)

Final preparation and closeout at KSC (e.g. lab support for sample/payload preparation, clean room)



Government Furnished Transportation Services (Cont.)



The following services are being proposed to be the responsibility of Transportation industry that will be provided with the GFS Transportation:

Service
Transportation out of Blast Danger Area (BDA)
Day of Launch Site Security
Landing & Recovery Team (NOTE: NASA representatives are included)
Crew Rescue & Return Team (NOTE: NASA representatives are included)
Landing Site Security (water and Land landing); NASA will provide US Coast Guard coordination

The following services are being proposed to be the responsibility of Destination Provider and are not provided by GFS:

Service *
Health Stabilization Plan (HSP) implementation
Quarantine Capabilities and Suit Room
Quarantine Security

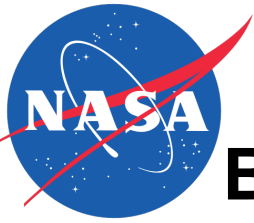
*NOTE: * NASA may offer some of the above services at GTA*



Additional Areas of Collaboration



- **NASA is also evaluating options to partner with industry on critical Destination systems**
 - Goal is to maximize Schedule and Technical risk reductions
- **Options being considered:**
 - NASA provides technical experts (SMEs) to support DDT&E
 - SMEs work inline and jointly develop system design w/ CLD provider
 - NASA provides resources, conduct joint trades, support DDT&E milestones (PDR, CDR, testing), etc.
 - May support a secondary goal to define areas of overlap with M2M systems and take advantage of synergies
 - NASA/Commercial Provider partner to jointly design critical systems
 - NASA build and deliver critical systems to integrate into CLD Provider's system design. Two major categories:
 - Hardware that can be clearly scoped to enable incorporation into a Provider's integrated system design
 - Would require development of an ICD to define interfaces and scope verification
 - Examples: Toilet, VIS, ECLSS components
 - High risk systems (e.g. OGA, UPA, Exercise) with goal to accelerate closed loop ECLSS availability



Business and Commercialization Plan Evaluation Approach



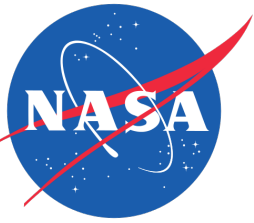
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Clarifications on CLDS Requirements / Assumptions



- **Due to the commercial nature of the CLDS contract. Financial stability is of paramount importance to NASA as well as the growth of a robust commercial LEO ecosystem with one of many customers. In light of that, NASA will be requesting in depth financial data on:**
 - Business Development progress in capturing non-NASA markets
 - Portfolio of products and services to be demonstrated and offered, including timing and pricing strategy
 - Status of signed contracts and agreements with subcontractors, vendors, suppliers, partners, and investors.
 - Year end financial reports using same format provided to investors/shareholders
 - Forecasted plan for the next year, including expected supplier and investor commitments, contracts, revenue, investments, financing, expenditures, obligations, and variance recovery plans
 - Regular summary of financial data and cash flow analysis



**Contract Approach:
Contract Type, CLINs, Period of Performance, Min-Buy**



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Proposed Contract Type & Competition Approach



The following Contract Acquisition Strategy is being considered for the CLDP Phase 2 Contract (CLDC):

- **Competition under FAR Part 15 – Contracting by Negotiation**
- **Utilize Full and Open Competition**
 - not a sole source
 - not a small business set-aside
- **Multi-award, Firm Fixed Price (FFP) - Indefinite Delivery Indefinite Quantity (IDIQ) hybrid contract.**
 - FFP Core CLIN for Certification
 - IDIQ Task Orders utilized for Mission Service Orders, Pre-Priced Additional Services, and Special Studies
- **Allow for an On-Ramp Clause to make additional awards in the future under the same RFP structure**
- **Strategy that allows for multiple awards (if funding is available)**
- **Evaluating need to protect for a “Leader/Follower” ATP approach**
 - Leader: Receives Award and is given ATP right away
 - Follower: Receives Award but ATP may be delayed



Contract Line-Item Number (CLIN) Structure Proposal



- The CLDC CLIN Structure is contemplated to include:
 - Firm Fixed Price (FFP) Core Contract for the CLD Certification (CLIN 1) with FFP Indefinite Delivery Indefinite Quantity (IDIQ) Task Orders for Mission Services Orders, Additional Services and Special Studies (CLIN 2-4)

CLIN	CLIN Category	Contract Type	Title	Price
1	Core	FFP	CLIN 1 - CLDS Certification Total Price (SUB-CLINS 1A-1C)	\$OFI
1A	Core	FFP	SUB-CLIN 1A System Certification	\$OFI
1B	Core	FFP	SUB-CLIN 1B Element-Capability Certification Sub Total Price	\$OFI
1B-1	Core	FFP	1B-1: Element #1 (ATP Required)	\$OFI
1B-2	Core	FFP	1B-2: Element #2 (ATP Required) - if proposed	\$OFI
1B-3	Core	FFP	1B-3: Element #3 (ATP Required) - if proposed	\$OFI
1B-4	Core	FFP	1B-4: Element #N (ATP Required) - if proposed	\$OFI
1C	Core	FFP	SUB-CLIN 1C-1 Crewed Flight Test Demonstration	\$OFI
2	IDIQ	FFP	CLIN 2 - CLDS End to End Services	TBD
3	IDIQ	FFP	CLIN 3 – CLDS Pre-priced Additional Services	TBD
4	IDIQ	FFP	CLIN 4 - CLDS Special Tasks & Studies	TBD

- NASA is still evaluating how to accommodate the certification of new visiting vehicles in the CLIN structure



CLIN 1 Proposal Overview

- **CLIN 1 of the CLD Contract will cover the Certification of the Contractors CLD, which is broken into three (3) major parts, all with entrance and exit payment milestones to satisfy completion of each CLIN.**
- **CLIN 1A: System Certification of the CLD**
 - CLIN 1A occurs from the contract effective date through the final Delivery Milestone for CLIN 1A: System Certification Review (SCR), which is for the Integrated CLDS to support a Full Operational Capability (FOC).
- **CLIN 1B: Element Certification**
 - CLIN 1B occurs from ATP/Kickoff through the end of the DDTE/Certification, which concludes the final Delivery Milestone for CLIN 1B: Uncrewed Demonstration.
- **CLIN 1C: Crewed Flight Test Demonstration**
 - CLIN 0001C occurs from ATP/Kickoff through the final Delivery Milestone for CLIN 1C: Crew Checkout and On-Orbit Demonstration

CLIN	CLIN Category	Contract Type	Title	Price
1	Core	FFP	CLIN 1 - CLDS Certification Total Price (SUB-CLINS 1A-1C)	\$OFI
1A	Core	FFP	SUB-CLIN 1A System Certification	\$OFI
1B	Core	FFP	SUB-CLIN 1B Element-Capability Certification Sub Total Price	\$OFI
1B-1	Core	FFP	1B-1: Element #1 (ATP Required)	\$OFI
1B-2	Core	FFP	1B-2: Element #2 (ATP Required) - if proposed	\$OFI
1B-3	Core	FFP	1B-3: Element #3 (ATP Required) - if proposed	\$OFI
1B-4	Core	FFP	1B-4: Element #N (ATP Required) - if proposed	\$OFI
1C	Core	FFP	SUB-CLIN 1C-1 Crewed Flight Test Demonstration	\$OFI



CLIN 2-4 Proposal Overview



- **CLIN 2-4 of the CLD Contract will cover the Fixed Price End-to-End Mission Services Orders, Additional pre-priced Services and Special Studies.**
- **CLIN 2: CLDS End to End Services**
 - The missions can be priced to three different configurations (IOC,CC, FOC).
 - Each of the three configurations have baseline set of requirements
 - With Payment Milestones, including a Mission Success Milestone
- **CLIN 3: CLDS Pre-priced Additional Services**
 - NASA is contemplating a list of Additional Services that the offeror can propose to and pre-price, which will be added onto the Mission Service Task Order price (CLIN 2) on an increment-by-increment basis
 - Additional Services may include: Additional Up/Downmass, Stowage, Payloads, Data Storage, Use of Crew Time, etc.
- **CLIN 4: CLDS Special Tasks & Studies**
 - NASA may require special studies to be performed or require labor hours to support unique activities.
 - The contract would require fully burden labor rates to be proposed, which would be used to develop special studies task order pricing.

CLIN	CLIN Category	Contract Type	Title	Price
2	IDIQ	FFP	CLIN 2 - CLDS End to End Services	TBD
3	IDIQ	FFP	CLIN 3 – CLDS Pre-priced Additional Services	TBD
4	IDIQ	FFP	CLIN 4 - CLDS Special Tasks & Studies	TBD



Proposed Period of Performance



- **Period of Performance:**
 - 10 Year Base Contract
 - One (1) Five Year Option Period
 - 15 Year total period of performance if option is exercised.
 - Considering increasing to Two (2) Five (5) Year Option Periods, total of 20-year POP
- **Period of Performance Assumptions:**
 - Build, Certification and Demonstration of a new LEO Destination likely to take 3-5 years
 - NASA desire is to issue IDIQ Mission Service Orders for at least 10 years after Certification completed to provide both government and industry return on investment
 - NASA desire to protect for ability to on-ramp additional providers over time
 - On-Ramp Selection may take 1-2 years to evaluate and award
- **NASA is not levying a requirement on the CLD Service Life, although it is implied the CLD must be available for the entire period of the performance including options.**
 - NASA to provide best practices associated with ensuring service life extension analysis is possible in the future.



Proposed Minimum-Buy Approach

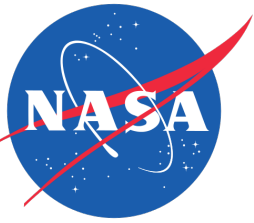


- **NASA recognizes a minimum-buy guarantee provides a level of certainty required for industry to secure investments and raise outside capital to ensure the CLD certification and services are not significantly dependent on the Government.**
- **NASA's approach to provide certainty, is to propose the following contract structure :**
 - CLIN 1A-C: System Certification thru Demonstration, as Firm Fixed-Price (FFP) with interim and final milestone payments.
 - CLIN 2: FFP IDIQ Mission Service Orders – which will include a minimum number of Mission Service Orders (Increment)
 - NASA is still evaluating the minimum number of IDIQ Mission Service Orders but is contemplating Six (6) Mission Service Orders (Increment)



- **On-Ramp Clause**

- The purpose of the Indefinite Delivery Indefinite Quantity (IDIQ) on-ramp is to ensure competition exists for future requirements not currently on contract by allowing qualified new service providers the opportunity to provide services.
- In accordance with this clause, the original solicitation (as amended or revised) shall remain open throughout the ordering period. The decision to request proposals under this clause will be solely at NASA's discretion and will only occur after the action has been synopsisized by posting to the Governmentwide Point of Entry (GPE). When requested, Offerors will be allowed to submit proposals that may result in contract award(s) to new providers and/or the addition of new capabilities to existing contracts.
- Upon award of each additional contract, the Government shall notify all present Contractors of the award, and the new Contractor(s) shall thenceforth be eligible to compete with all present Contractors for the award of future IDIQ task orders.



Procurement Strategy



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Proposed Evaluation Approach



- **The purpose of this section is to share a draft of the evaluation factors planned for the CLDC Request for Proposal and potential evaluation weighting.**
 - *(Price Evaluation and Past Performance details are excluded from this presentation)**
- **Topics:**
 - Evaluation Factors Order of Importance (Riddle)
 - Price/Mission Suitability/Small Business
 - Mission Suitability Discriminators
 - Mission Suitability Weighting



Draft Evaluation Order of Importance (Riddle)



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- **NASA approach for proposal evaluation is expected to consist of 3 major factors:**

<u>Price</u>	<u>Mission Suitability</u>	<u>Past Performance</u>
Pricing Approach	Technical Approach	Past Performance Evaluation
Pricing Template Fill-Ins	Management Approach	
	Small Business Approach	

- **NASA expects to evaluate the Mission Suitability and Past Performance factors, when combined, to be approximately equal to the Price factor.**



Mission Suitability Proposal Discriminators



- Within the Mission Suitability, the RFP outlines 3 standard subfactors. NASAs approach is to identify specific discriminators the proposal will be evaluated on:
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Price	Mission Suitability	Past Performance
1. Pricing Approach	1. Technical Approach	3. Past Performance Evaluation
2. Pricing Template Fill-Ins	2. Management Approach	
	3. Small Business Approach	

Tentative Approach

Technical Approach	Management Approach	Small Business Approach
1. System Architecture & Capabilities Integrated Element Architecture; Assembly Sequence; ConOps; Capabilities	1. Program Management Organization Structure; Key Personnel; Crew Code of Conduct; Risks	1. Small Business Subcontracting Plan Plans to meet subcontracting goals
2. Certification & Engineering Approach Cert Plan; Demo Plan; IOC/CC/FOC; Standards and Compliance; Human in The Loop; System Reliability	2. Commercialization Plan Investments; Cash Flow	2. Commitment to Small Business Program Demonstrate commitment to small business
3. Services & Operations Approach Integrated Master Schedule; Mission Integration & Ops, Utilization Integration & Ops; GSE/GFE; Multi-User Approach; Maintenance; Transportation	3. Insight & Approvals Plan Approach to meet Insight and Approval Plan	
4. Safety & Mission Assurance Approach to meet S&MA	4. Work Plan Milestone plan approach	



Mission Suitability Weighting Approach



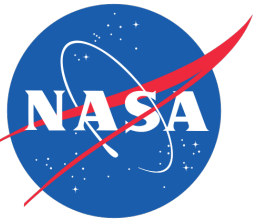
- The RFP will also weight each subfactor of the Mission Suitability which will derive a points score for each proposal. Based on the draft subfactors on the previous page, weighting and proposal page count is expected to be in the range of:

Price	Mission Suitability	Past Performance
1. Pricing Approach	1. Technical Approach	3. Past Performance Evaluation
2. Pricing Template Fill-Ins	2. Management Approach	
	3. Small Business Approach	

Mission Suitability Subfactors	Weighting
1. Technical Approach	TBD
2. Management Approach	TBD
3. Small Business Approach	TBD
Total	1000

- NASA tentatively expects that the Technical subfactor will hold slightly more or equal weight to the Management subfactor, with the Small Business subfactor holding lower weight than the others.

All information is considered draft and subject to change prior to final RFP release.



CLD Jurisdiction and Export Control



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- **NASA desires to increase certainty that U.S. courts have jurisdiction to hear a criminal cases associated with activities on a CLD in order to provide both NASA and CLD Provider a stronger foundation to ensure crew compliance with requirements on-orbit**
- **Concern:**
 - While on the CLD, a criminal offense could occur by a CLD crew member without clear jurisdiction for prosecution by a U.S. court
- **Proposed Approach:**
 - NASA to require all modules where U.S. Government astronauts are required to have access to be registered in the U.S.
 - For foreign-launched modules registered in the U.S. and conducted by a foreign entity, the CLD Provider will identify how liabilities for damage associated with the launch will be covered
 - Based on CLD Provider's CONOPS, an extradition treaty may be required to obtain personal jurisdiction of crew members returning to non-U.S. landing sites
 - NASA to require flow down of requirements to all crew members placing them on notice that while on the CLD they are within U.S. jurisdiction (for all modules registered in the U.S.)
 - NASA to require flow down of the Crew Code of Conduct to all crew members, placing them on notice of expected standards of behavior



- **NASA desires to increase certainty that U.S. courts have jurisdiction to hear a civil cases associated with activities on a CLD in order to provide both NASA and CLD Provider a stronger foundation to ensure crew compliance with requirements on-orbit**
- **Concern:**
 - People or property on a CLD may cause harm to other people or property on board, resulting in possible civil claims between individuals and entities (including the United States Government (USG)); uncertainty may exist regarding which jurisdiction has personal and subject matter jurisdiction over such claims
- **Proposed Approach:**
 - NASA to require all modules where U.S. Government astronauts are required to have access be registered in the U.S.
 - NASA to require flow down of the Crew Code of Conduct to all crew, putting them on notice of expected standards of behavior
 - NASA to require CLD Provider to require crew members to agree that CLD activities are governed by US law and to submit to the exclusive jurisdiction of United States courts
 - NASA will require (TBD) insurance including coverage for claims resulting from disputes between parties on a CLD (consistent with proposed overall liability approach)



Export Control



- **Background:**

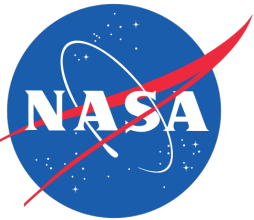
- Export control considerations for commercial space stations are crucial to ensure compliance with U.S. regulations
- NASA does not own the CLD but is a main customer

- **Concern:**

- As CLDs become operational, they will involve international collaboration, technology transfers, and foreign personnel, increasing the risk of unauthorized exports of controlled technologies

- **Proposed Approach:**

- NASA will require Contractor/CLD Provider to comply with standard (i.e., Federal Acquisition Regulation-based) export control clauses in the contract and be responsible for export licenses
- The commercial provider is responsible for export control and ensuring required licenses for its team members in conducting international activities and discussions
- NASA will be responsible for any licenses required for NASA exchange of export-controlled data



Data Rights



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- **Purpose**

- Propose Overview of Data Rights and recommendation for CLD Contract (CLDC)

- **Agenda**

- Proposed: Gov't Rights for Provider Produced and/or Delivered Data Hierarchical Layout of Rights in Data
- Miscellaneous Definitions (To be proposed in contract language)
- Proposed Data Rights Approach for CLDC
- Backup



Proposed Gov't Rights for Provider Produced and/or Delivered Data



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- The Government's rights in data in FAR-based contracts can range from ownership coupled with possession to finite, limited access.
- Possible data rights/license categories include:

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- (A) **Special Works Rights** - The Gov't receives delivery coupled with "unlimited rights" or title by assignment (ownership) from the contractor in data identified as "Special Works." The Special Works clause is primarily used for the production or compilation of data (other than limited rights data or restricted computer software) for the Government's own use, or when there is a specific need to limit distribution and use of the data or to obtain indemnity for liabilities that may arise out of the content, performance, or disclosure of the data.
- (B) **Unlimited Rights** –The rights of the Gov't to use, disclose, reproduce, prepare derivative works, distribute copies to the public, and perform publicly and display publicly, in any manner and for any purpose, and to have or permit others to do so.
- (C) **Government Purpose Rights** - The rights of the Gov't to use, modify, reproduce, *manufacture*, release, perform, display, or disclose data within the Government without restriction; and release or disclose data outside the Gov't and authorize persons to whom release, or disclosure has been made to use, modify, reproduce, manufacture, release, perform, display, or disclose that data for U.S. Government purposes. [Deviation]
- (D) **Limited Rights** – The rights of the Gov't in limited rights data is set forth in the Limited Rights Notice of the contract. Limited Rights Data includes trade secrets or commercial or financial and confidential or privileged. Limited Rights Data may be reproduced and used by the Gov't with the express limitation that they will not, without written permission of the Contractor, be used for purposes of manufacture nor disclosed outside the Gov't; except that the Government may disclose these data outside the Gov't for specified purposes set forth in the Limited Rights Notice.
- (E) **Insight** – The general right of the Gov't to conduct inspections and be afforded access to data needed for monitoring of contractor quality and Government-identified metrics and contracted milestones, and it may also involve the review of contractor work procedures and records. At a minimum, access to the "Insight" data is the ability for the Government and its support service Contractor personnel, both remotely and onsite at the Contractor's facilities, to locate and review all "Insight" data associated with areas of Insight set forth in the contract, such as the PWS, in a usable and readable format. Insight is a continuum that can range from low intensity, such as reviewing quarterly reports, to high intensity, such as performing surveys and reviews. Insight will be defined in section H.

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Miscellaneous Definitions (To be proposed in contract language)

- **Insight Data** – Data associated with areas of Insight in the performance of this contract as defined in unique H Insight Clause
- **Research Data** - Data that is CLDC contractor generated on the CLD Platform in the performance of this contract for USG Payloads and/or USG Experiments (timelines, procedures, training, commands, telemetry, images, edge computing analyses, etc.) that the Gov't may want to get title to and/or limit distribution and use of per the Special Work H Clause. This includes but is not limited to electronic data collected by individual investigations documenting the results of the sponsored experiment. This includes numerical values, images, videos, sound recordings, and the research sponsor hardware collects these data. **(Defined in CLDP-PLN-1122, Science Data Management Plan)**
- **“Media”** - For purposes of this clause, the term “Media” shall include, but not be limited to, audio, imagery, video, virtual or augmented reality, or other works of mass communications produced by Contractor under the Contract.
- **CLDS Research Related Mission Data** – CLDS Data collected describing conditions in research segments of the CLDS including, but not limited to, temperature, humidity, time-resolved vehicle accelerations, magnetic field, acoustic, atmosphere composition, and radiation level. For the instruments collecting mission data, such data shall also include calibrations, coefficients, algorithms, instrument documentation, and any other ancillary information. **(Defined in CLDP-PLN-1122, Science Data Management Plan)**
- **CLDS Health & Status Data** – Operational Health and Status data collected is pertinent to understanding of the safe operations and performance of the CLDS Critical Systems, CLDS Resource and Interface Payload Data, and CLDC Contractor facilities, etc.



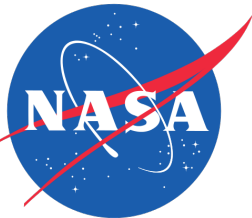
Proposed Data Rights Approach for CLDC

- NASA anticipates requiring, at a minimum, the following rights in specified sets of data produced or provided by the contractor. For example:

A. Special Works Rights	B. Unlimited Rights	C. Government Purpose Rights	D. Limited Rights	E. Insight Data
• “Media” shall include, but not be limited to, audio, imagery, video, virtual or augmented reality, or other works of mass communications produced by Contractor under the Contract • Research Data - Data that is contractor generated on the CLD Platform for USG Payloads and/or USG Experiments (timelines, procedures, commands, telemetry, images, etc.) that the Gov’t may want to get title to and/or limit distribution and use of the data to protect USG Researcher.(Defined in Science Data Mgmt. Plan)	• CLDS Research Related Mission data – aka, specific data that describes the CLD mission environment critical to understanding the research data results, which needs to be provided to research community via an open science archive platform. (Defined in Science Data Mgmt. Plan) • CLDS Health & Status Data	• Joint Design Data Cases where the development has been performed with mixed funding where the government has an investment in the design or interface data for future programs like Regen ECLSS, Docking ICDs, etc. This is only for further development or modification of data provided by the Government (but doesn't cover the Gov't data itself). • Government Task Agreements (GTA)	• All Deliverable Data such as DRDs and Milestone Review Data as defined in the contract. • Computer Restricted Software	• All Insight Data as defined in the contract. Detailed design data, MRB data etc. Frequency and viewing rights established.

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Less Rights to Govt



Crew Code of Conduct



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Proposed Approach for Crew Code of Conduct



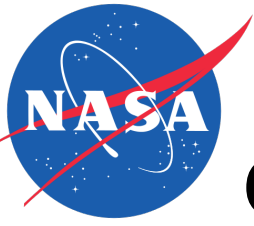
- The ISS Crew Code of Conduct (CCoC) is set forth in [Title 14 of the Code of Federal Regulations \(CFR\) Subpart 1214](#)
- Since 14 CFR 1214 is ISS-specific in content and requires an ISS governance framework, it can not be applied unmodified to the CLD Contract (CLDC)
- Modifications to 14 CFR 1214 will be required to apply a Crew Code of Conduct to CLDs
- The Program will determine minimum requirements to be included in the contract while seeking and taking into consideration industry feedback
- **Crew Code of Conduct (CCoC) will establish a set of behavioral standards for crewmembers of Commercial LEO Destinations (CLDs) to help:**
 - Protect the hardware, crew, and success of the mission
 - Maintain a cohesive operating environment
 - Protect the NASA and Contractor brand and assets
 - Protect the intellectual property of CLD participants
- **CCoC will apply to all crewmembers. Additional requirements for Government astronauts may be set by the Agency.**



Proposed Approach for Crew Code of Conduct (CCoC)



- **NASA will define minimum standards for CLD CCoC; NASA may identify best practices beyond the minimum**
- **NASA will require CLDC offerors to submit a CCoC in their proposals via a Type 1 DRD to cover the following areas:**
 - Applicability
 - Responsibilities of Crewmembers
 - General Rules of Conduct
 - Use of Position
 - Mementos and Personal Effects
 - Disciplinary Policy
 - How the CLDC crewmembers will be bound by the CCoC
 - Definitions of Terms
 - Physical and Information Security Guidelines
 - Protection of Human Research Subjects
 - Differences in requirements for government vs non-government astronauts, if any
 - Other contractor ideas
- **Related content that was covered in the ISS 14 CFR 1214.4 will be included in CLDC Section H, Special Contract Requirements:**
 - Roles, Responsibilities, and Required Experience for CLD Commander
 - Chain of Command and Succession On-Orbit
 - Relationship Between the Destination Commander (On-Orbit Management) and the Flight Director (Ground Management)



Crew Selection and Certification Approach Overview



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Crew Selection Approach



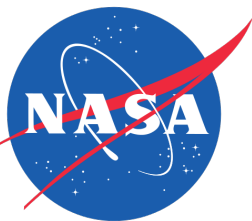
- **NASA envisions working jointly with the commercial provider(s) to establish a set of criteria for the selection of CLD flight crews.**
 - NASA will maintain crew selection criteria and responsibility for flight assignments for the NASA astronaut corps but will meet CLD established criteria.
- **Proposed Approach to Crew Selection Criteria**
 - NASA would specify minimum requirements/standards
 - Minimum age
 - Background check requirements including criminal history, major financial issues, etc.
 - Minimum set of medical and psychological qualifications
 - NASA approval requirement for crews from Designated Countries Tier II and III
 - Additionally, NASA will identify best practices for standards above and beyond the minimum
 - Commercial Provider will submit a Crew Selection Process DRD to NASA for review and approval



Crew Certification for Flight Approach



- **Training: Crew must have received role-specific training according to the intent of standards defined CLDP-STD-1150 Commercial Low-Earth Orbit Development Program (CLDP) Operations Standards**
 - Commercial provider(s) will perform an assessment of crew readiness based on crew performance observed in training
- **Medical and Behavioral: The commercial provider will perform assessment of crew of behavioral health, psychological suitability, and crew interpersonal dynamics according to evaluation criteria based upon the concepts in CLDP-PLN-1111 Crew Health Ops Con (CHOC) and the intent of standards defined in and CLDP-STD-TBD Commercial Low-Earth Orbit Development Program Medical Standards and Capabilities.**
 - NASA will provide the commercial provider with data demonstrating that NASA crew meet or exceed medical and behavioral standards defined in CLDP-STD-TBD Commercial Low-Earth Orbit Development Program Medical Standards and Capabilities.



Liability Framework



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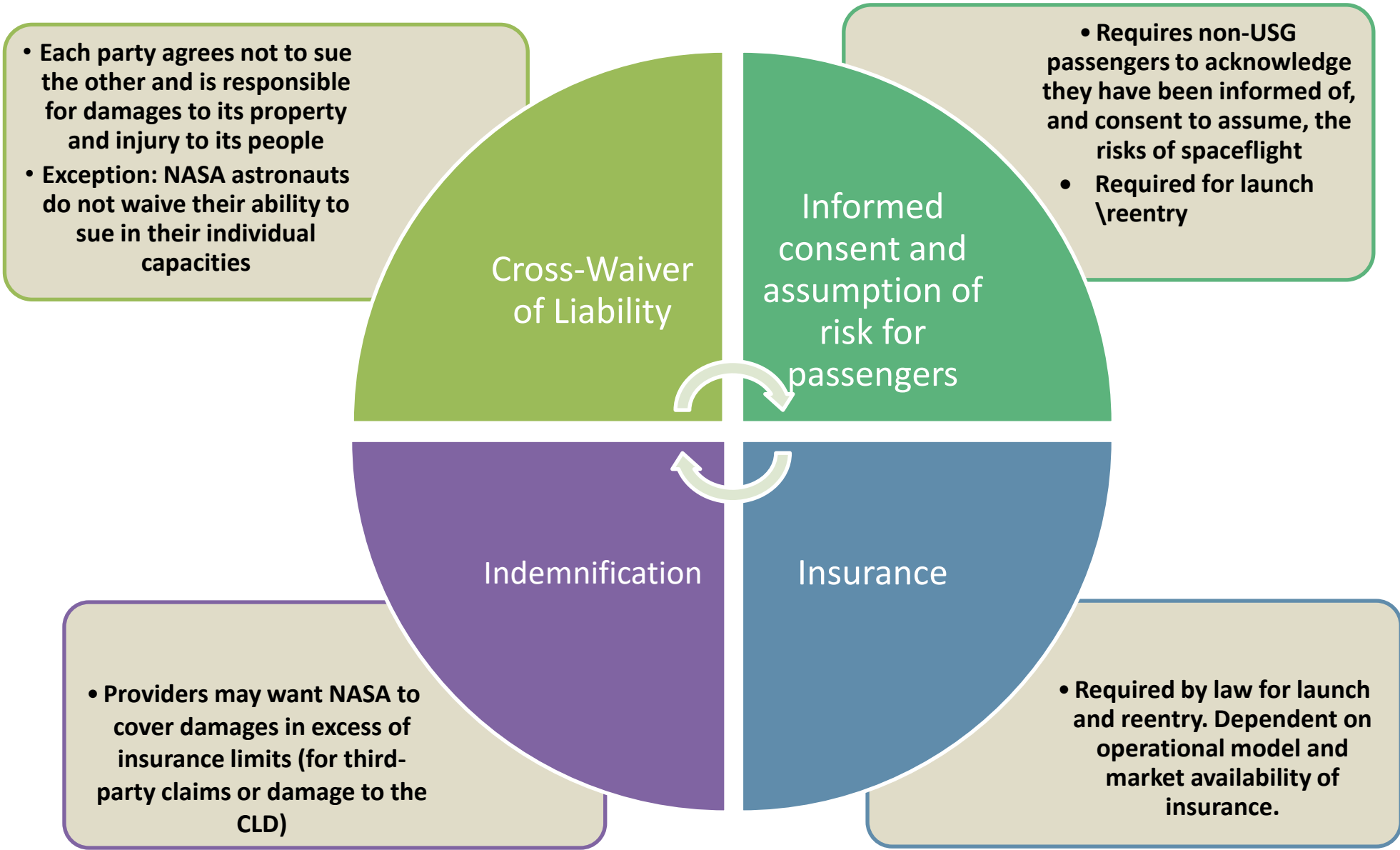


Overview



- There is no legally required liability framework for commercial on-orbit activities
- There is a legal framework for FAA-licensed launch\reentry, which will apply to transportation to\from the Commercial Low Earth Orbit (LEO) Destination (CLD)
- The proposed approach would establish a liability framework for the CLD contract informed by:
 - Other recent human spaceflight contracts (e.g., Commercial Crew, PAMs)
 - Laws and regulations that govern FAA-licensed commercial launch and reentry
- The objective of the liability framework is to address risk between NASA and the CLD and transportation providers in a way that shares responsibility, mitigates litigation, protects the U.S. Government while enabling commercial viability, and protects Government Astronauts
- There are four main components to the liability framework:
 - Cross-waivers
 - Informed consent
 - Insurance
 - Indemnification

CLD Phase 2 Contract Liability Framework





Cross-Waivers of Liability



Background:

- Under a cross-waiver, each party waives its right to sue the other for injury or death of its personnel, loss or damage of its property, and any other damages (except in the case of willful misconduct and other exceptions)
- NASA regulations require cross-waivers for space exploration activities involving a launch, and for ISS activities
- NASA cross-waiver applies to activities defined as “Protected Space Operations” and has been tailored for recent contracts

Proposed Approach:

- Cross-waiver is broad, covering all CLD contract activities
- Cross-waiver applies between NASA and CLD Provider; between NASA and transportation providers; and between the CLD Provider and transportation providers
- Contractors flow down the waiver to their “Related Entities” (e.g., subcontractors, partners, customers) requiring them to waive claims against the U.S. Government (USG) and USG’s “related entities.” Private astronauts also waive claims.
- The CLD Provider decides whether to require waivers between itself and its own “related entities”
- USG astronauts are performing official duties and are not required to waive personal legal rights

Takeaways:

- Reduces all parties’ potential liability and risk of litigation.
- Each party is responsible for repairing its own property, without recourse to recover if damage is caused by another party. This risk may be mitigated by contractor insurance and potential USG indemnification of the CLD itself.



Informed Consent\Assumption of Risk



Background:

- Spaceflight participants on FAA-licensed launches are legally required to state in writing that they have been informed of and agree to assume the risks of the launch. There is no regulatory requirement for on-orbit activities.
- NASA has included this requirement as a condition of its contract allowing commercial flights of non-NASA astronauts to the ISS for commercial purposes (Private Astronaut Missions, or “PAMs”)

Proposed Approach (similar to PAMs):

- Require CLD Provider to obtain written informed consent\assumption of risk statements from non-USG astronauts, including:
 - Inform users of hazards and risks associated with use of the CLD;
 - Inform users that NASA has not certified the CLD for other than U.S. Government (USG) use; and
 - Inform users that the CLD Provider, not NASA, is responsible for operation of the CLD and for non-USG activities on the CLD
- Astronauts who are USG employees will not be required to sign informed consent, because they are performing official duties

Takeaways:

- Allows a participant to make an informed decision about their participation before waiving their rights
- Mitigates risk of liability for the Government and the CLD Provider
- Consistent with FAA license requirements for transportation to and from the CLD



Background:

- Insurance protects the policy-holder and other insureds if a covered event occurs, and provides compensation for injury\ damage to third parties
- Insurance for injury\damage to third parties, and for damage to USG property, is required by law for FAA-licensed launch and reentry activities. There is no legal requirement for insurance on-orbit. NASA has required various types of insurance for spaceflight contracts such as PAMs and Commercial Crew.
- NASA has no legal authority to provide insurance for CLDs
- NASA Astronauts will be on CLDs with Private Astronauts and may conduct operations for the CLD Provider or its customers. NASA also may provide equipment on the CLD for the CLD Provider\Private Astronauts' use.

Proposed Approach:

- Require insurance covering third-party claims for injury or property damage
- Require insurance for injury\damages the CLD Provider's Private Astronauts cause to USG astronauts or property
- Require CLD Provider to insure its property, if NASA indemnifies for loss of the provider's property
- Determine amount of insurance based on input from insurance brokers and industry

Takeaways:

- Provides financial recourse for third parties and the CLD Provider
- Cross-waiver and potential indemnification may mitigate contractor's financial risk for damages that exceed insurance.



Background:

- CLD Providers have indicated that loss of the CLD, damages to third parties, and the limited availability of insurance are significant considerations in their ability to develop and operate a CLD
- Indemnification allows the Government to agree in advance to pay claims for injuries or damages arising from the contract, which the contractor otherwise would be liable for, when such claims exceed contractor insurance
- The Government can only indemnify when it has specific statutory authority to do so. Public Law 85-804 indemnifies contractors for “unusually hazardous activities” but can be used only when the contract has a nexus to facilitating the national defense
- Indemnification covers damage to contractor and Government property as well as third-party claims. Indemnification requires approval by the Administrator.

Proposed Approach:

- Require insurance to the extent it is available
- Consider whether indemnification could be authorized
- Indemnification for transportation to/from CLD will be under FAA launch license (51 U.S.C. § 50915)

Takeaways:

- Would cover on-orbit “unusually hazardous” activities that are within the scope of the contract. It would not cover claims arising from commercial activities of the CLD provider, which are outside the scope of the contract.
 - Provides financial protection for third parties